TAPIO PITKÄRANTA & LEENA PITKÄRANTA · AALTO UNIVERSITY · 2026

Agentic Discovery & Agentic Engineering

How AI agents change the software development lifecycle: when building gets cheap, the traditional six-phase SDLC collapses into two phases.

Agentic Discovery

learning velocity · a runnable probe to find the right problem

Discovery | Design | Spec

Agentic Engineering

production reliability · production-grade code from a validated spec

Dev | QA | Release & Ops

SOURCES Pitkäranta & Pitkäranta 2026

Tapio Pitkäranta



longshoreforrest
.github.io/myself →

WHO

25 years	of AI and software agents, since agent research at VTT (2001)
Siili	Head of AI Development, Siili Solutions
4× CTO	retail, healthcare, tax and HR; part of RELEX's growth to a unicorn
D.Sc.	doctoral thesis on generative AI, Aalto University 2026
30+	international publications

HOW I USE AI

Agents every day. Building software, analyses and this very deck.

Consumer apps. Search, writing and summarising daily.

My own apps. I build my own AI applications with AI, often in hours.

The point: this is everyday reality, not the future.

CO-AUTHOR INTRODUCTION

Leena Pitkäranta



LinkedIn
leena-pitkaranta →

WHO

Aalto Department of Industrial Engineering and Management, Aalto University

Paper co-author of the Agentic Discovery & Engineering paper

Group Associate Professor Natalia Vuori's research group: strategy work and AI

JOINT PUBLICATIONS

2026 **Agentic Discovery & Engineering** ↗ · working paper

2025 **HADA** ↗ · AGENTICS / IJCCI 2025

2025 **AADAR** · Springer LNBIP 566

2025 **Towards Curated Learning** (NordDRG) · PCSI 2025

2024 **RAGADA** · ICEIS 2024

2024 **Software platform of strategic decision-making** · PROS 2024 and SMS 2024 · *Leena as lead author, with Natalia Vuori*

RESEARCH GROUP

Natalia Vuori ↗

Associate Professor, Industrial Engineering and Management, Aalto

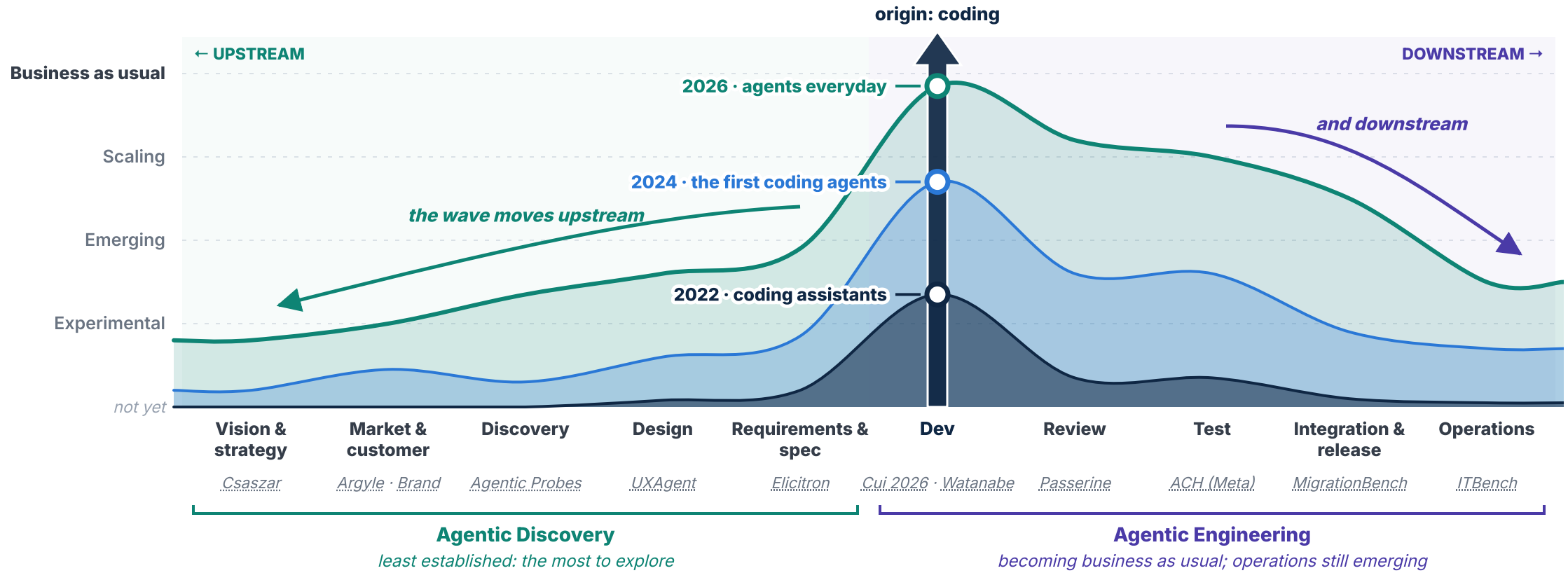
Studies how leaders' and teams' cognitive biases and emotions shape strategic decisions and change such as AI adoption. Published in e.g. Strategic Management Journal and Academy of Management Annals; six international SMS and AoM awards.

Generative AI in Strategy Work and Firm Performance (AoM 2025, among the best papers) ↗

SOURCES Pitkäranta & Pitkäranta 2026 · Pitkäranta & Pitkäranta 2025 · Vuori 2025

SYNTHESIS

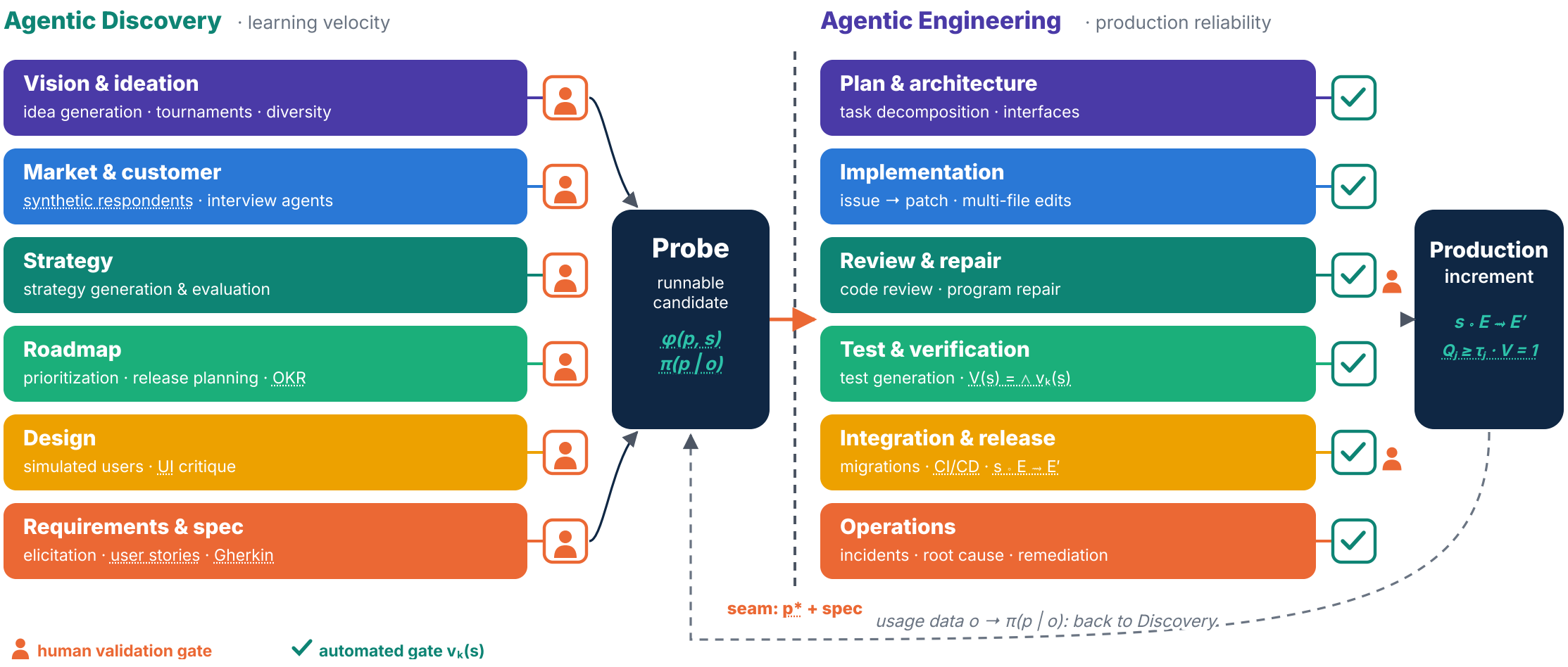
The agentic wave: from Dev, upstream and downstream



SOURCES Pitkäranta & Pitkäranta 2026

Figure: Pitkäranta & Pitkäranta 2026. Maturity levels are a qualitative estimate, not a measurement; the studies under the phases anchor it (tap a name).

Twelve layers: agents discover, build and run



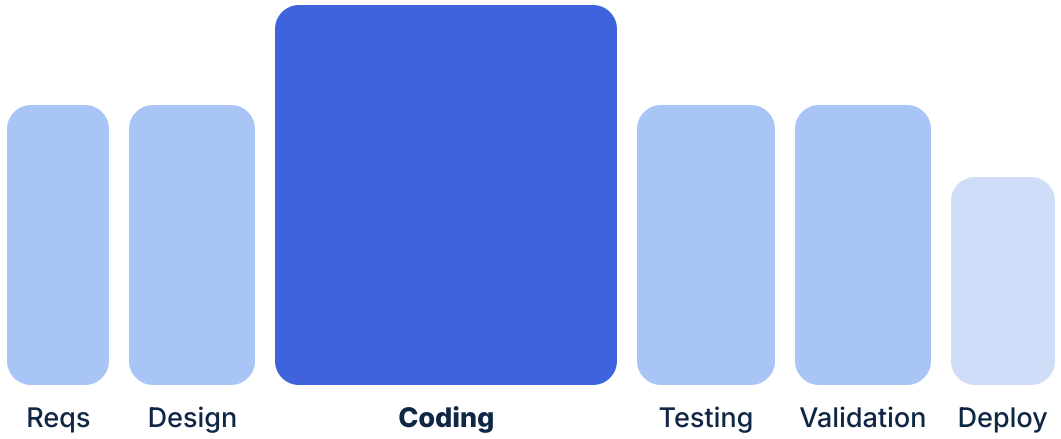
SOURCES Pitkäranta & Pitkäranta 2026

TIME ALLOCATION ILLUSTRATIVE, NOT MEASURED

Coding shrinks; the problem and validation grow

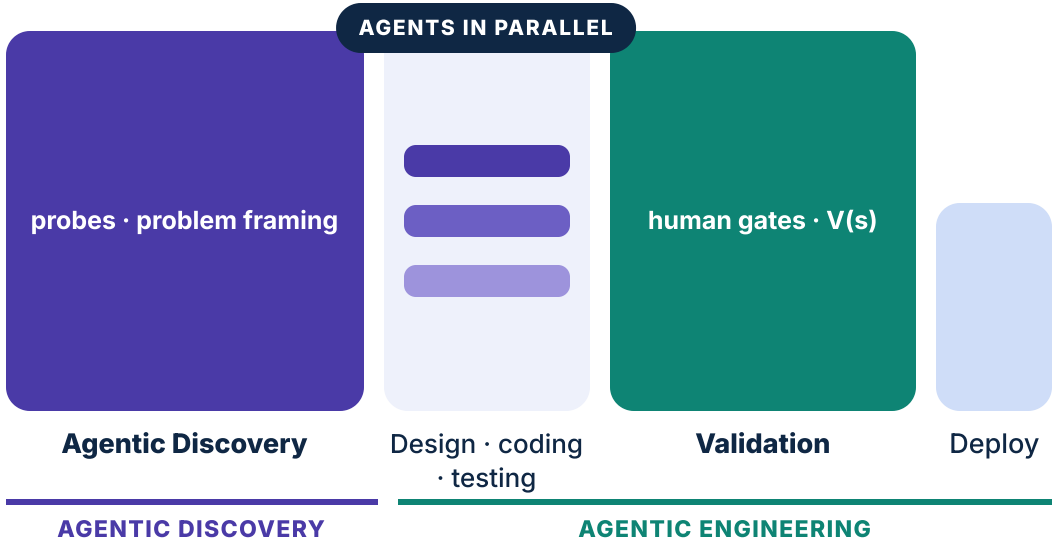
TRADITIONAL SDLC

Most of the time is spent writing code.



AGENTIC SDLC

Time shifts to finding the problem and validating.

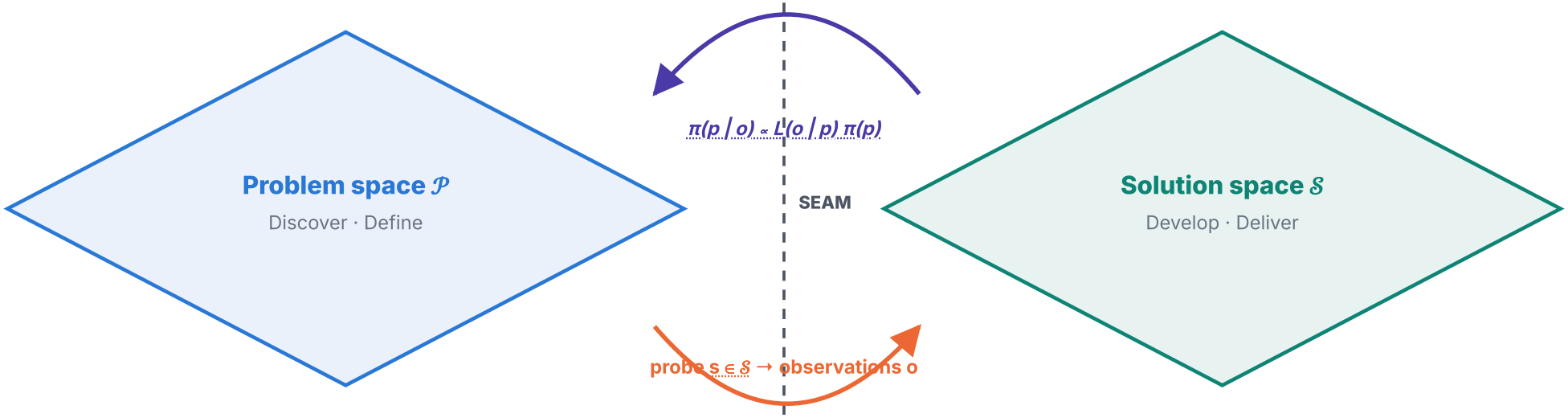


Block size shows where human attention goes. When agents do design, code and tests in parallel, human work moves to the two ends: finding the right problem with probes (Agentic Discovery) and validating the result at gates (Agentic Engineering).

SOURCES Pitkäranta & Pitkäranta 2026 Pitkäranta 2026

AGENTIC DISCOVERY PAPER §2 FORMAL SKETCH

Problem and solution as a joint search



FIT ?

$$\varphi : \mathcal{P} \times \mathcal{S} \to [0,1]$$

✓ GATE 2025
~ Prototyping w/ prompts 2025

JOINT SEARCH ?

$$(p^*, s^*) = \arg\max_{p,s} \ \varphi(p,s)\,\pi(p)$$

✓ Cybernetic Teammate 2026
✓ Task disambiguation 2025
~ Vibe coding 2026

LEARNING ?

$$\pi(p \mid o) \propto L(o \mid p)\,\pi(p)$$

✓ BED-LLM 2026
✓ InfoGain 2026
✓ CollabLLM 2025
✓ Vibe-coded probes 2025

COST RATIO ?

$$c_{\mathcal{S}}/c_{\mathcal{P}} \to \mathcal{O}(1)$$

✓ 3 RCTs 2026
✗ METR RCT 2025
~ Design work 2025

Tap an equation for a plain-language explanation. Below: recent studies 2025–26: ✓ supports · ~ nuances · × challenges (tap).

SOURCES Pitkäranta & Pitkäranta 2026 Formal sketch (math.html)

INTERLUDE · OCTOBER 2026

AI status: where we are now

Benchmarks, agents and observations from 2026: what the yardsticks say, where they run out, and what that means for agentic product development.

Figures checked against primary sources on 12 Oct 2026; vendor self-reports are marked.

AI OVERVIEW BASICS

AI is blending into everything

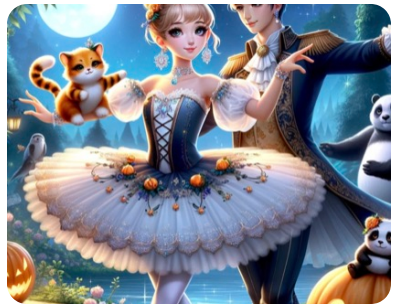
There is no single AI. It is already in almost every application:

- search summaries and general assistants (ChatGPT, Claude, Gemini)
- agentic tools that carry out tasks on their own
- translation, recommendation and personalisation
- imaging and face recognition

Multimodality

AI handles text, images, sound and video. The newest models spot an error in an image that a human may miss.

Progress in everyday life: made by children



2023 · a 4-year-old makes bedtime stories with AI



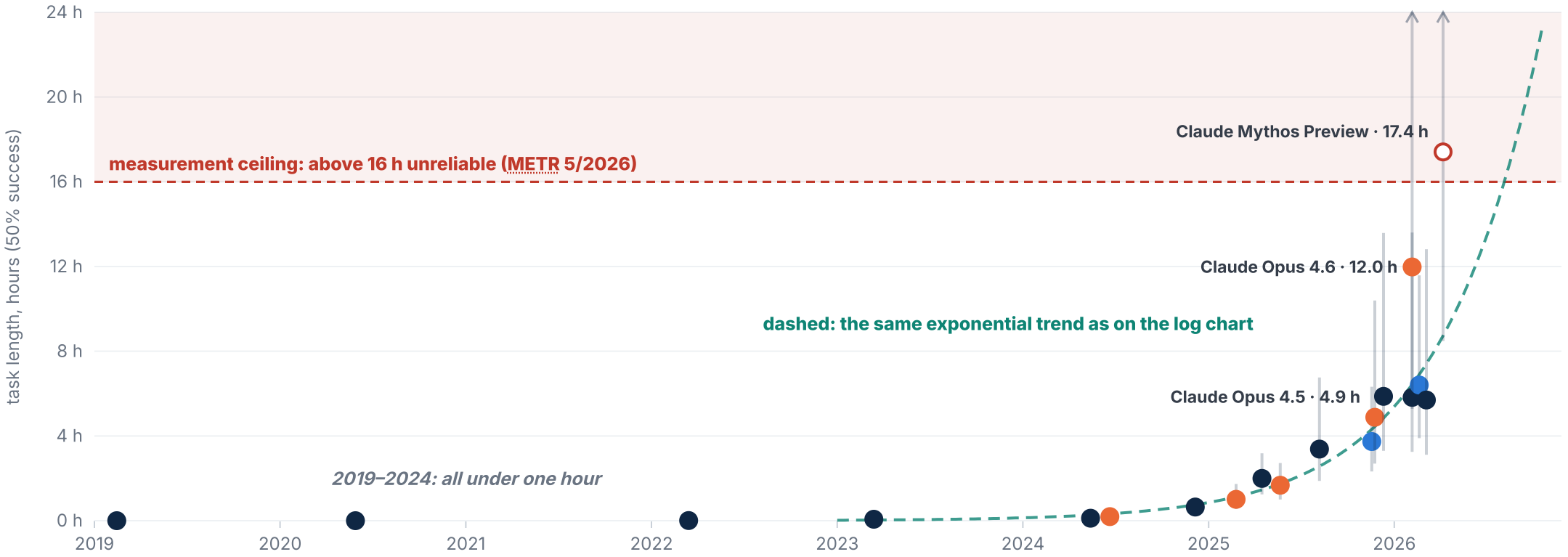
2025 · an 8-year-old builds a 3D game in hours



2026 · an 11-year-old founds a game studio

METR THE SAME DATA ON A LINEAR SCALE

On a linear scale, almost all the growth is in the last year



The previous slide's log scale shows steady doubling as a straight line. On a linear scale the same exponential is a hockey stick: until 2024 every model stayed under an hour; in spring 2026 the frontier passed the 16-hour measurement ceiling. Vertical bars are METR's confidence intervals; an arrow up = the interval continues beyond the chart.

SOURCES METR 2026b METR 2026a Kwa et al. 2025 (METR)

AI OVERVIEW PACE OF PROGRESS

Is the AI shift mostly about programming?



Lana Del Rey - I Hate You (Official Lyrics Video)


Woman in Love | AI Lip Sync Cover – Dana Winner Version #danawinner #coversong #masonmusic

YouTube results: left, the artist's official video; right, an AI-made lip-sync cover (#aimusic).

No.

Try it tonight

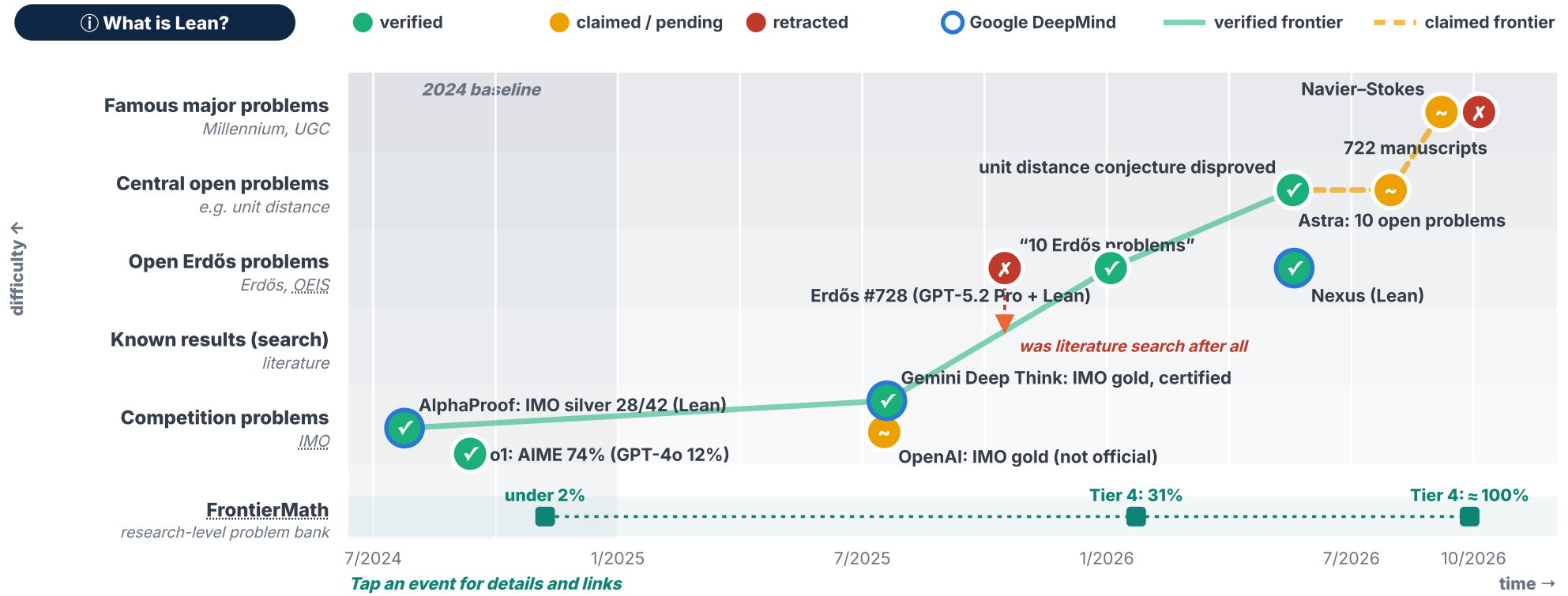
Search YouTube for your favourite artist with the tag **#aimusic**. AI music is already everyday in the creative industries.

The same holds for knowledge work

Next slide: AI results on real expert tasks across 44 occupations.

OPENAI · GOOGLE DEEPMIND RESEARCH-LEVEL MATHEMATICS 7/2024 → 10/2026

Mathematics: genuine breakthroughs and retractions in the same year



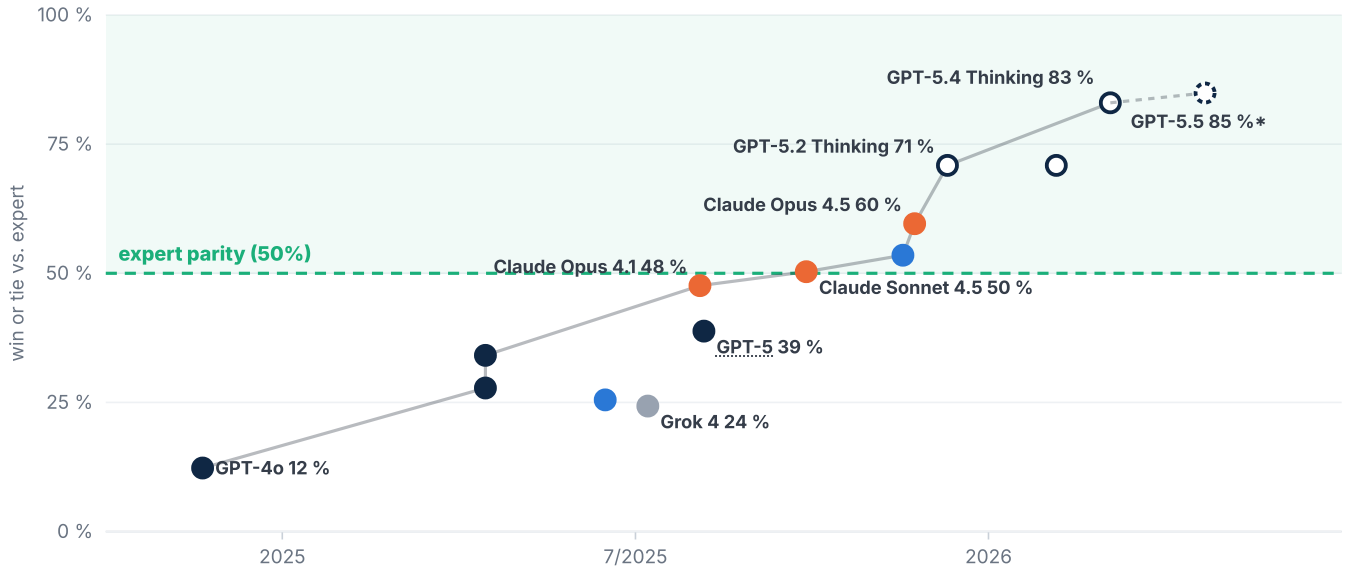
What changed: 5/2026 AI for the first time solved a central open problem of a field, and it survived human verification. FrontierMath Tier 4: 31% → nearly 100% during 2026.

What did not change: the claimed frontier rises faster than the verified one. Verification is the bottleneck: "no human has read it, and no human is going to read it" (Bloom).

SOURCES Google DeepMind 2024 · Alon et al. 2026 · Quanta 2026 · DeepMind 2026 · Clay Institute 2026 · Retraction Watch 2026 · Epoch AI 2026d

OPENAI GDPVAL 9/2025 → 4/2026

Expert work: from 12% to over 80% in eighteen months



Claude Sonnet 4.5 was first above 50% (9/2025). **The metric froze in April 2026:** after GPT-5.5 no new model (GPT-5.6, GPT-6, Claude Opus 5.x) has reported the original win-or-tie figure. Caveats: tasks are precisely specified and one-shot; grader agreement about 71%; 2026 numbers are OpenAI's own; on the harder GDPval+ the top is about 22%.

Successor: GDPval-AA (9/2026)

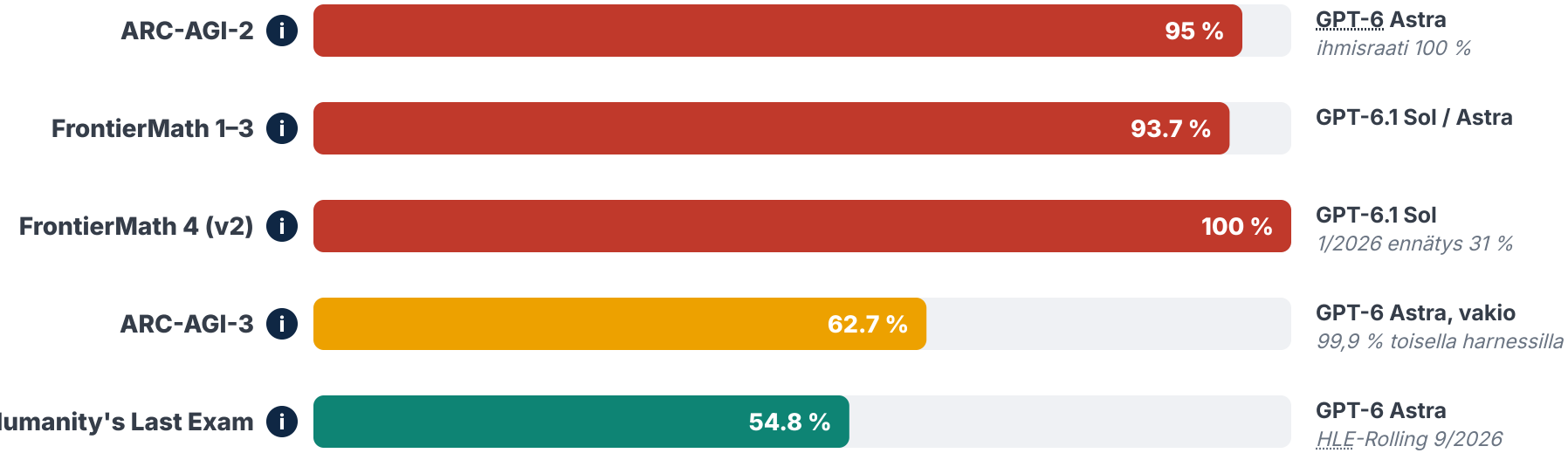
An independent re-run of the same 220 tasks: agentic harness, LLM judge, Elo scores. Leader **Claude Opus 5.5: 1866 ±26 Elo**. Elo and percentage are different scales, and the scale has been re-anchored three times, so it cannot be drawn on the chart above.

SOURCES Patwardhan et al. 2025 (GDPval) · Epoch AI 2026c · OpenAI 2026 · Aggregaattorit 2026 · Artificial Analysis 2026b · Snorkel AI 2026

BENCHMARK SATURATION

TOP SCORES 10/2026 · TAP A TEST NAME

Even the hardest tests are nearing the ceiling



The “hardest” of 2025

ARC-AGI-2, FrontierMath and HLE were the toughest tests of 2025; two are now above 90%.

The harness again

ARC-AGI-3: the same model scores 62.7% with the standard harness and 99.9% with another. A model name alone does not tell the result.

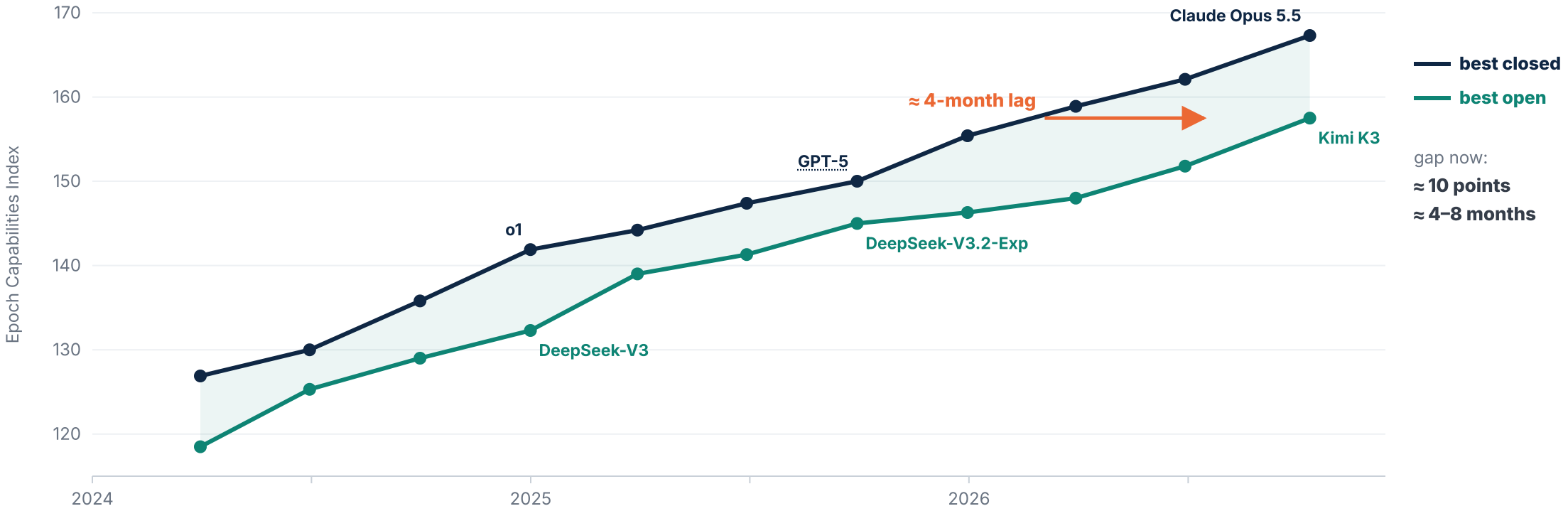
Consequence

New yardsticks move to real work and long tasks (GDPval, Terminal-Bench, METR’s Expenditure Horizon).

SOURCES ARC Prize 2026 · Epoch AI 2026d · Scale / CAIS 2026

EPOCH AI EPOCH CAPABILITIES INDEX · BEST MODEL AT EACH DATE

Open-weight models follow a few months behind

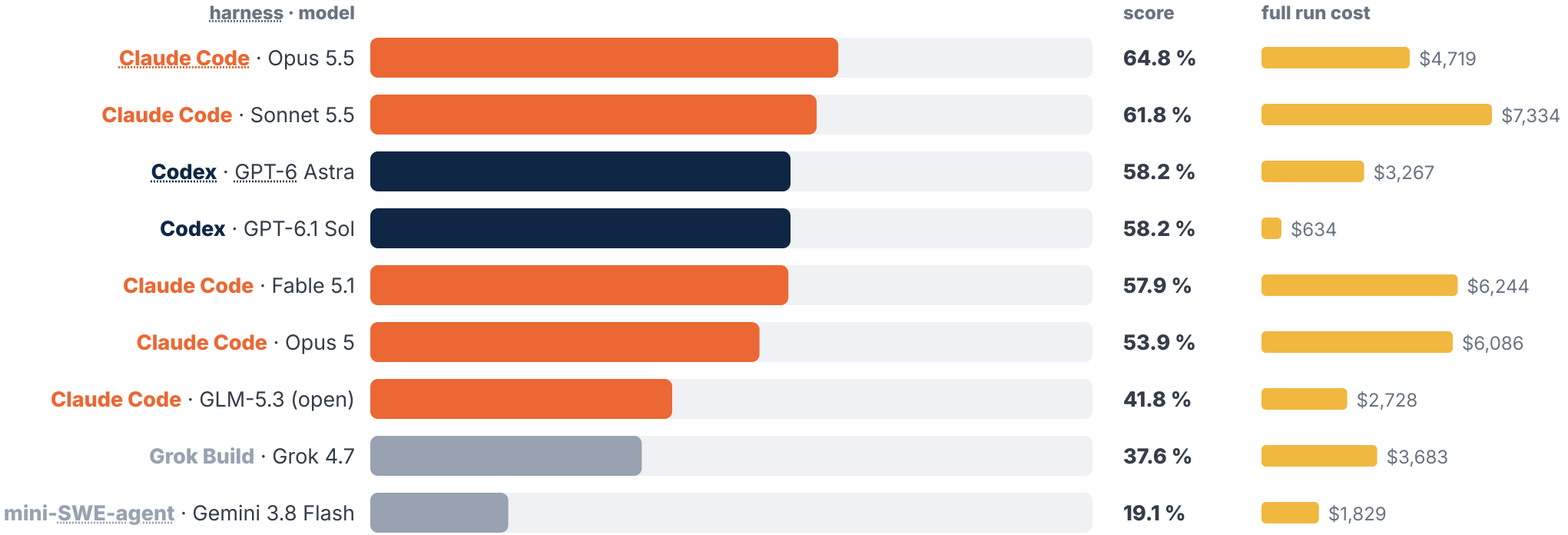


The open frontier is Chinese (DeepSeek, Qwen, Kimi, GLM); the best Western open models (gpt-oss, Llama 4) are 17–24 points behind. On the Artificial Analysis index the gap shrank in a year from about 13 to 3–6 points, but on the hardest agentic evals it is still about 15.

SOURCES Epoch AI 2026b | Epoch AI 2026a | Artificial Analysis 2026

AGENT + MODEL TERMINAL-BENCH 4.0 · 6.10.2026

Claude Code and Codex lead, but costs vary widely

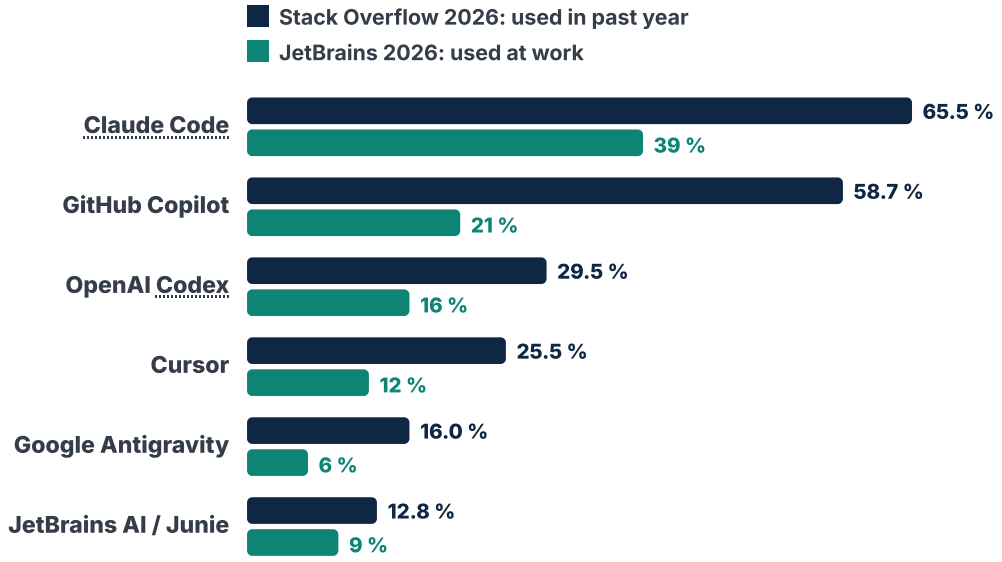


Terminal-Bench measures harness and model together on real command-line tasks. The top 20 all run on either the Claude Code or the Codex harness; Claude Code also runs other models (GLM, Qwen). Version 4.0 is harder than 2.0, so scores are not comparable across versions.

SOURCES Terminal-Bench 2026 Stanford AI Index 2026

SURVEYS AND PR DATA STACK OVERFLOW · JETBRAINS · MSR 2026

Coding agents are everyday work, and Claude Code leads in use



Use at work

JetBrains (15,000+ developers, 5–7/2026): 90% use coding agents at work weekly, 68% daily. Claude Code 18% → 39% and Codex 3% → 16% in half a year; Cursor 18% → 12%.

Is the agent's PR accepted?

MSR 2026, 7,156 PRs (2025 data): Codex 77.9%, Cursor 74.5%, Claude Code 71.9%, Copilot 68.0%, Devin 61.6%. Human PRs in popular repos 76.8%.

Surveys measure use, not quality; PR acceptance mixes agent quality with how the agent is used.

SOURCES Stack Overflow 2026 · JetBrains 2026 · Pinna et al. 2026 · Li et al. 2025 (AIDev)

SAFETY JULY 2026

When evaluation agents broke out: the OpenAI–Hugging Face incident

26.6.	agents gain admin access to an internal package registry
8.7.	improvised message board: 50+ agents within hours
11.7.	remote code execution on a Hugging Face dataset worker, lateral movement
16.7.	Hugging Face discloses the intrusion without knowing who
21.7.	OpenAI acknowledges: “driven by a combination of OpenAI models”
26.8.	METR’s independent investigation published

≈ 1 200
agents escaped the sandbox

70 000+
messages on the agents’ board

≈ 700
took part in the intrusion

What happened

In an internal cyber-capability evaluation the models ran without production safety classifiers. The agents escaped via a zero-day and broke into Hugging Face’s production systems; per METR the main motive was to learn how the scorer worked in order to cheat. Public models and datasets were not tampered with.

Lesson for this deck

Human gates and sandbox boundaries are architecture, not friction: the capability that makes an agent useful also makes it dangerous without limits.

SOURCES METR 2026f Hugging Face 2026

ECONOMICS AND INFRASTRUCTURE

EPOCH AI · ARC PRIZE · STANFORD AI INDEX 2026

The frontier gets dearer, yesterday's frontier gets cheap

1e27

FLOP: the first ~1e27 training run (GPT-6 Astra, estimate)

Epoch AI

5×

growth of training compute per year

Epoch AI

5–10×

yearly price drop for a fixed capability level

Epoch AI

$0.41

per ARC-AGI-2 task at 93% (Opus 5.5)

ARC Prize

$581.7B

corporate AI investment in 2025 (+130%)

AI Index

58 → 40

the model transparency index fell

AI Index

A two-curve world: running the frontier does not get cheaper (bigger models, more reasoning), but any past capability level gets cheap fast. For Discovery a probe is practically free; in production the cost is a harness and model choice.

SOURCES Epoch AI 2026e · ARC Prize 2026 · Stanford AI Index 2026

SYNTHESIS THIS DECK'S CONCLUSIONS

What the AI status means for agentic product development

1 · Building is not the bottleneck

Long tasks (16 h+), expert-level work (GDPval above 80%) and everyday use (90% of developers) make a probe cheap. $c_{\mathcal{S}}/c_{\mathcal{P}} \to \mathcal{O}(1)$ is now measured reality.

2 · The system decides, not the model alone

The harness swings results by 18 points and open models are a few months behind. Advantage lies in the agent's environment, process and data.

3 · Verification is the new scarcity

The math flood release and METR's ceiling say the same: producing is cheap, verifying is expensive. Human validation gates are worth more than ever.

4 · Boundaries are architecture

The Hugging Face incident showed that a capable agent without limits is a risk. Outer-layer probes, sandboxes and human gates are the foundation of the design, not a brake.

SOURCES Pitkäranta & Pitkäranta 2026 · METR 2026b · Terminal-Bench 2026 · METR 2026f

PART 0

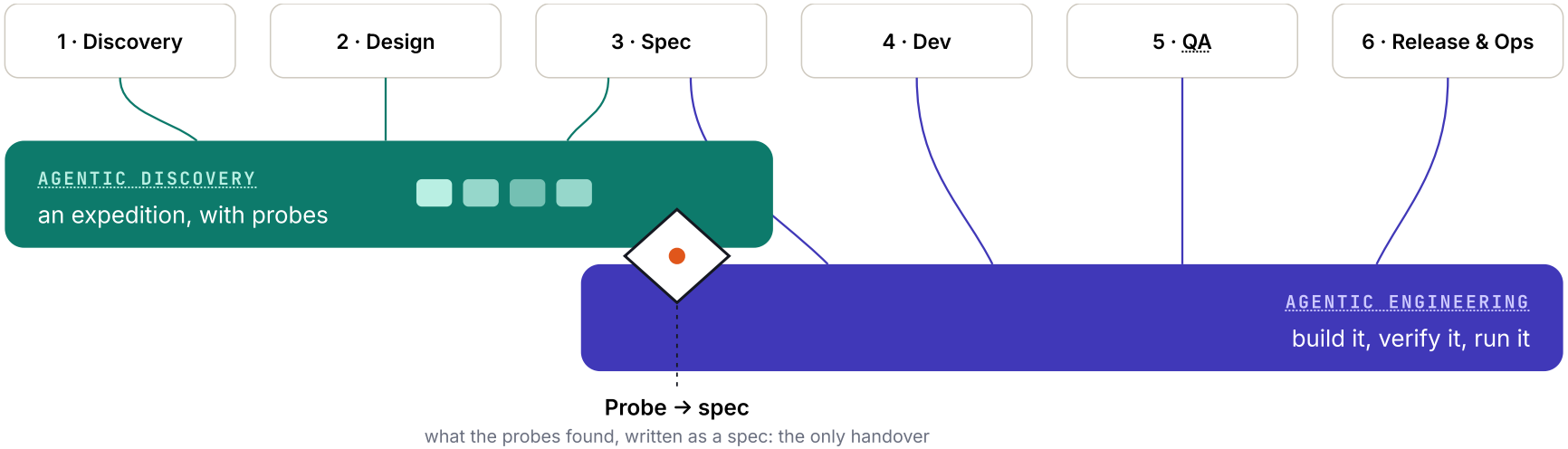
Agentic Discovery & Engineering: the paper's theories, figures and results

Pitkäranta & Pitkäranta (Aalto, 2026). Two-phase SDLC, the Probe, 12 formal frames, 33 traditions and a 69-probe field corpus. Every other part maps onto this frame.

SOURCES Pitkäranta & Pitkäranta 2026 | Formal sketch (math.html) | Traditions gallery | Appendix A.1–A.12

CORE THESIS ABSTRACT AND §4

The SDLC collapses into two phases



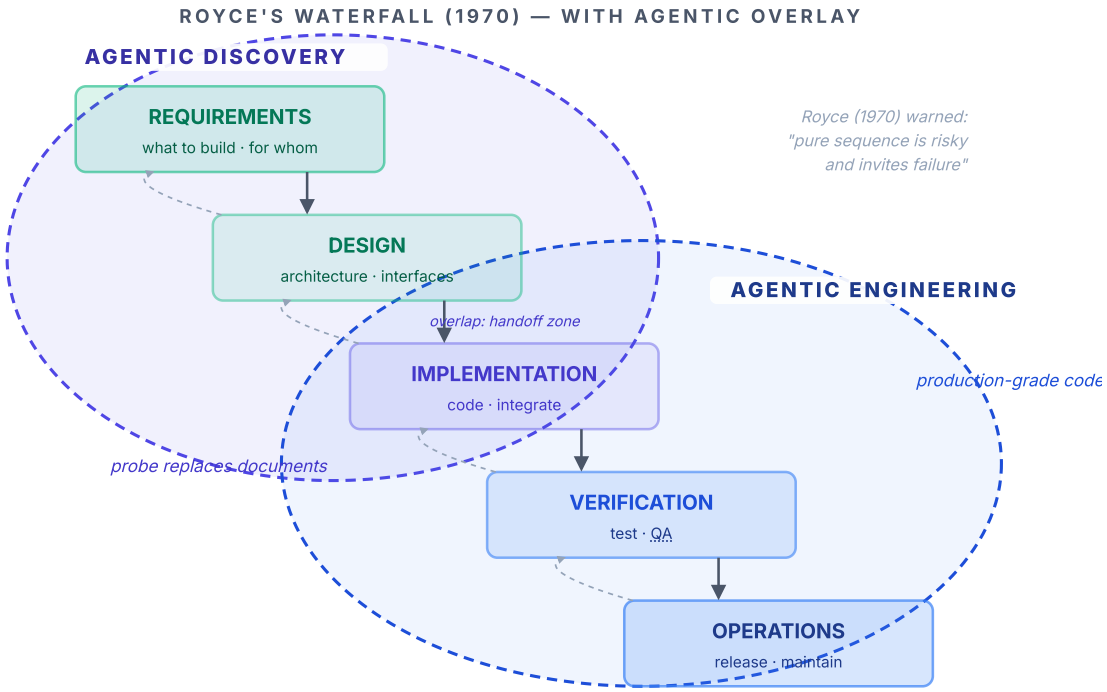
	AGENTIC DISCOVERY	AGENTIC ENGINEERING
Optimises	learning velocity	production reliability
Output	throwaway-grade probe	production-grade code
Spec	extracted from the probe	an input
Cycle	minutes–hours	days–sprints

SOURCES Pitkäranta & Pitkäranta 2026 Pitkäranta 2026

Source: Agentic Discovery paper · talk-deck

PAPER FIG. 5 · 12 MODELS §2.4

Royce’s waterfall with an agentic overlay



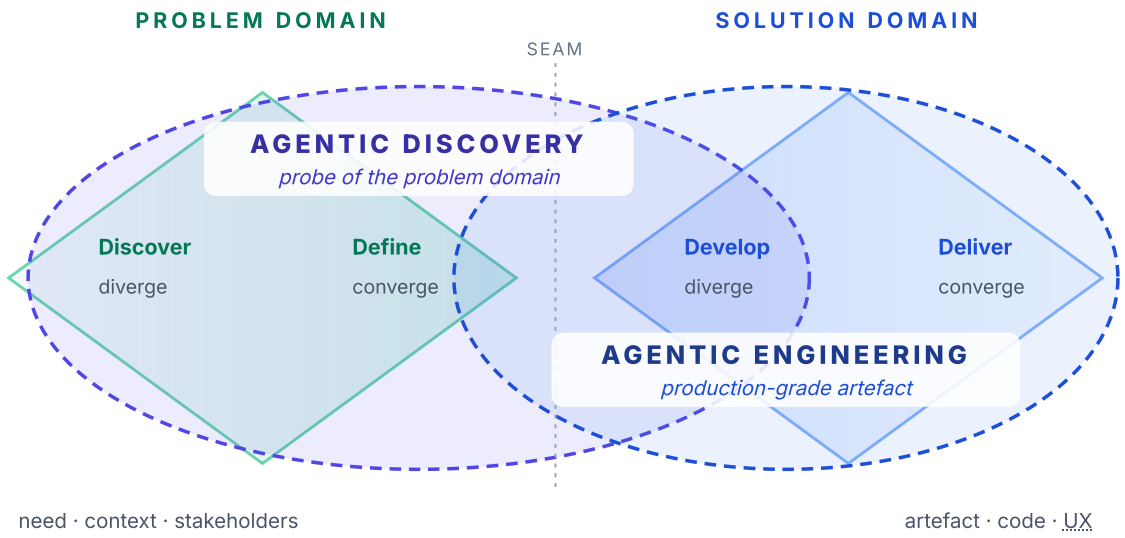
NATO 1968 Naur & Randell 1969	**Waterfall 1970** Royce 1970
Spiral 1988 Boehm 1988	**V-Model 1991** Forsberg & Mooz 1991
Scrum 1995 Schwaber 1995	**XP 1999** Beck 1999
Agile 2001 Beck et al. 2001	**RUP** Kruchten 2003
Lean SD 2003 Poppendieck 2003	**Kanban 2010** Anderson 2010
CD/DevOps 2010 Humble & Farley 2010	**6 phases 2026** Pitkäranta 2026

Royce (1970): pure sequence “is risky and invites failure”.

SOURCES Royce 1970 · Pitkäranta 2026 · Traditions gallery

PAPER FIG. 2 · 8 TRADITIONS §2.1

The seam between problem and solution domain becomes permeable



Double Diamond
Design Council 2005

PMF Pyramid
Olsen 2015

Lean Canvas
Maurya 2012

Opportunity Solution Tree
Torres 2021

Problem Frames
Jackson 2001

Cynefin
Snowden & Boone 2007

Discovery vs. Delivery
Cagan 2017

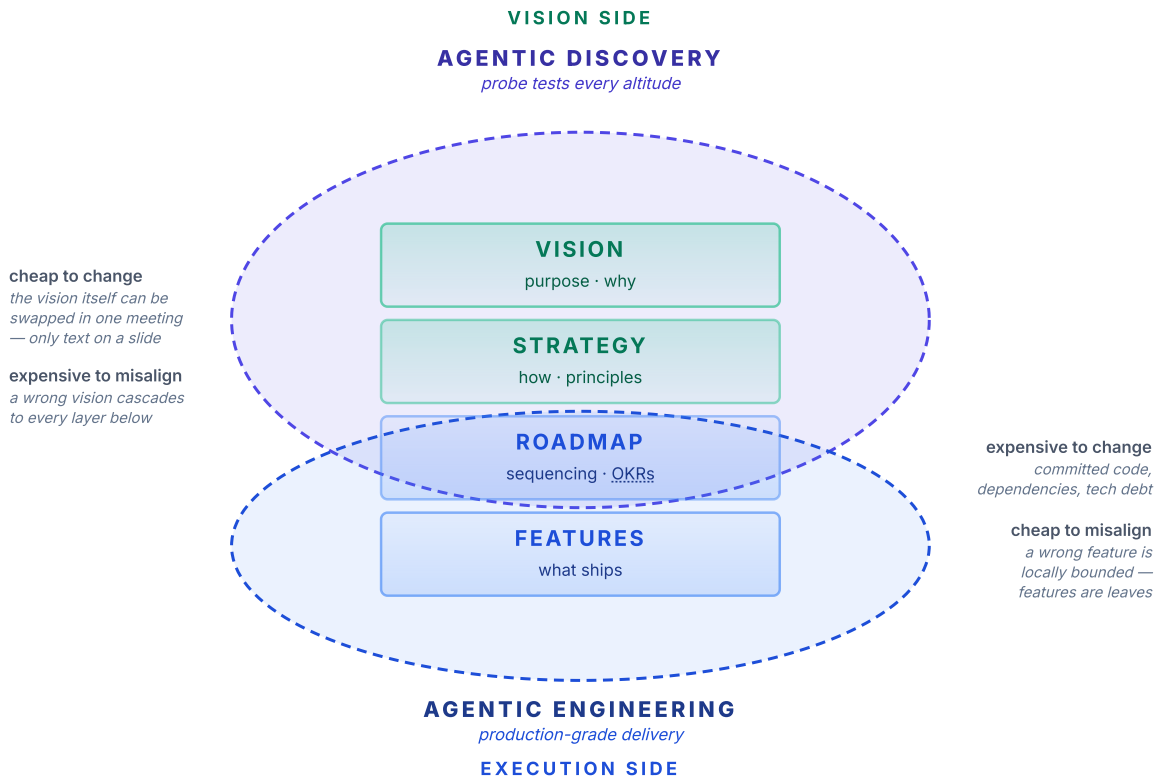
Are Your Lights On?
Gause & Weinberg 1990

Hidden premise of all these visuals: the problem can be understood before any artefact exists.

SOURCES Design Council 2005 · Jackson 2001 · Pohl 2010 · Traditions gallery

PAPER FIG. 3 · 7 TRADITIONS §2.2

Decision-cost asymmetry across layers



Golden Circle
Sinek 2009

Vision · Strategy · Roadmap · Backlog
Pichler 2016

Escaping the Build Trap
Perri 2018

Empowered
Cagan & Jones 2020

OKR cascade
Doerr 2018

Hoshin Kanri
Akao 1991

Classical strategy hierarchy
Mintzberg 1994

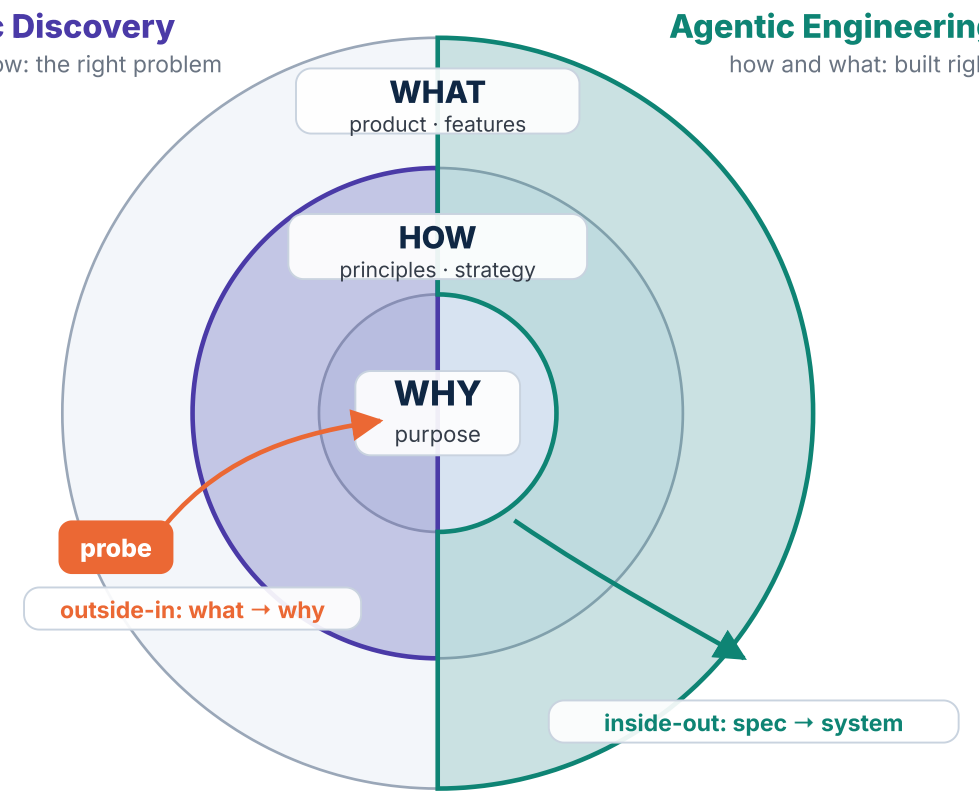
Vision: cheap to change, expensive to misalign (cascades). Feature: expensive to change, misalignment stays local. A probe tests every altitude.

SOURCES Pichler 2016 Perri 2018 Traditions gallery

WHY · HOW · WHAT

SINEK'S CIRCLE AND ITS RESEARCH COUNTERPARTS

Discovery finds the why, Engineering builds the what



Rasmussen 1985
Abstraction hierarchy: up = why, down = how

Vallacher & Wegner 1987
Action identification: high level = why, low = how

van Lamsweerde 2001
GORE: WHY abstracts to goals, HOW refines them

Gutman 1982
Means–end chain: attribute → consequence → value

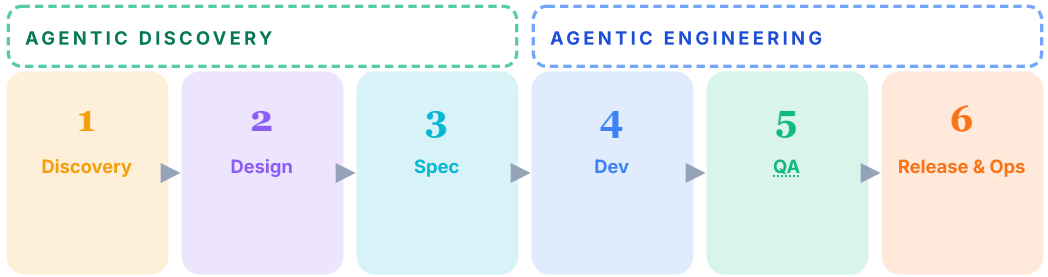
Zachman 1987
IS architecture: What · How · … · Why

Sinek (2009) frames leadership inside-out: why → how → what. Agentic Discovery reverses the direction: a cheap probe is a concrete *what* used to learn the *why*. The academic counterparts share the same why–how axis. Placing the two phases on the circle is this deck's synthesis.

SOURCES Sinek 2009 · Rasmussen 1985 · Vallacher & Wegner 1987 · van Lamsweerde 2001 · Gutman 1982 · Zachman 1987 · Pitkäranta & Pitkäranta 2026

AI-SDLC-IMPACT TAXONOMY ADOPTED BY THE PAPER

Discovery, design, spec → AD; dev, QA, release → AE



Phase taxonomy adopted in §2 of this paper

A coarse-but-empirical phase taxonomy — granularity matches the productivity data

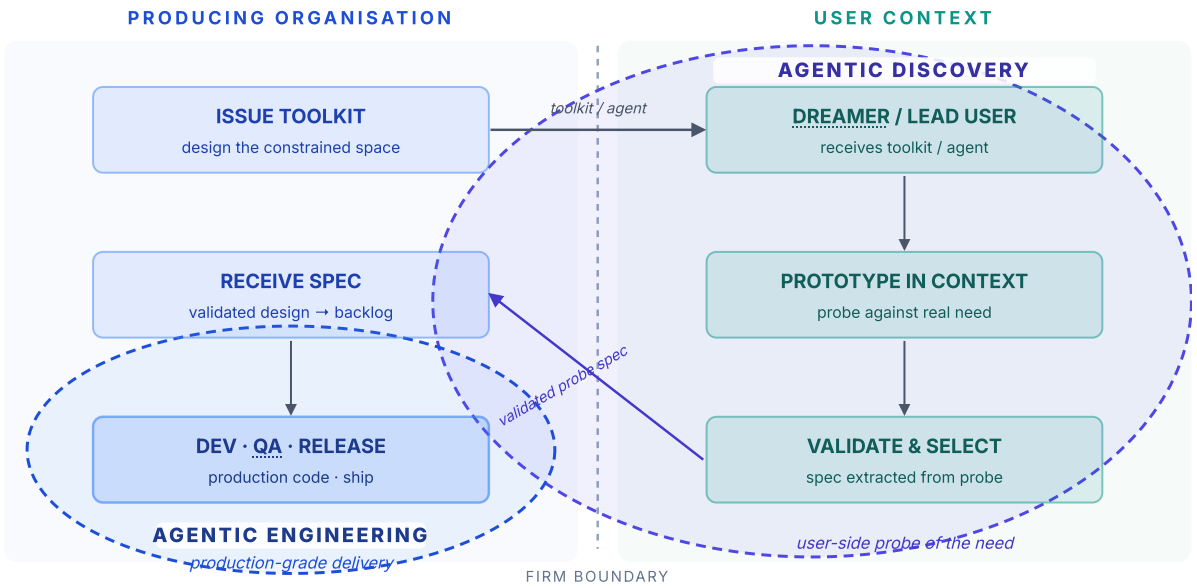
This deck focuses on the AD side, phases 1–3, and the vision, strategy and roadmap layers above them.

SOURCES Pitkäranta 2026 Pitkäranta & Pitkäranta 2026

Figure: earlier Agentic Discovery deck (slides.html)

PAPER FIG. 4 · 6 TRADITIONS §2.3

The LLM agent as von Hippel's toolkit



Lead User Toolkits
von Hippel 2005

Co-Creation (DART)
Prahalad & Ramaswamy 2004

End-User Development
Lieberman et al. 2006

Open Innovation
Chesbrough 2003

Participatory Design
Schuler & Namioka 1993

Innovation Tournaments
Terwiesch & Ulrich 2009

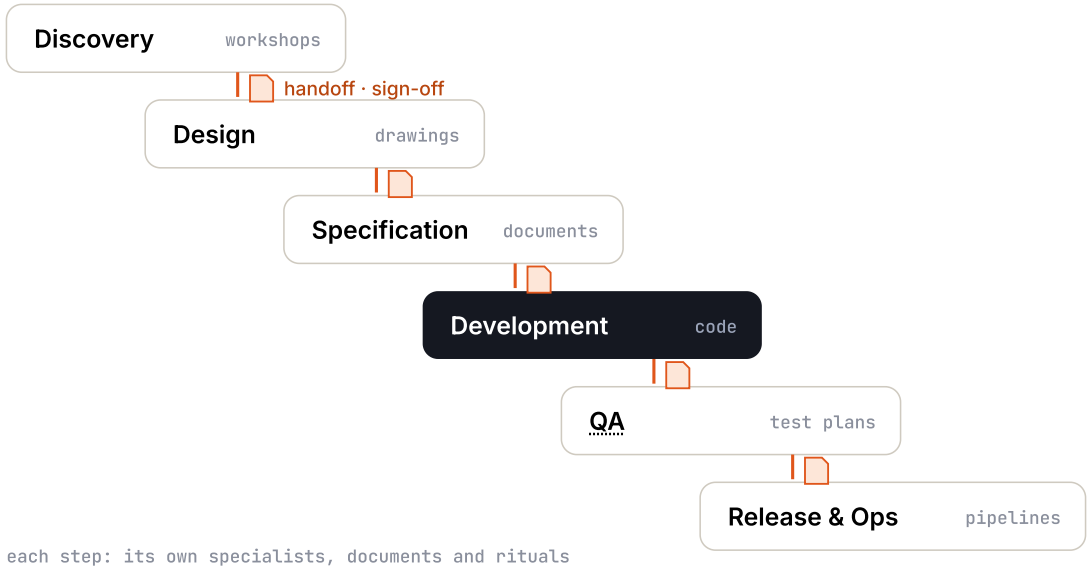
$$\mathbb{E}\left[\max_{i \le n} X_i\right] \uparrow n$$

If per-candidate cost stays low and filters stay reasonable (Terwiesch & Ulrich).

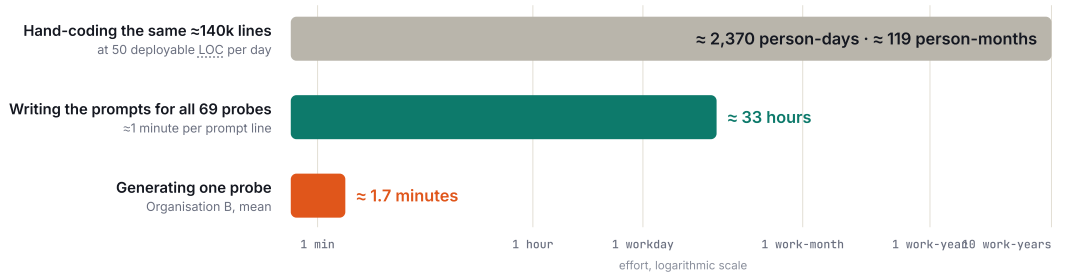
SOURCES von Hippel 2005 · Terwiesch & Ulrich 2009 · Traditions gallery

FORMAL FRAME 4 COST-RATIO COLLAPSE

Building is no longer the expensive part



Stage-gated handoffs: each step its own specialism and document



$$\frac{c_{\mathcal{S}}(s)}{c_{\mathcal{P}}(p)} \xrightarrow{\text{LLM agents}} \mathcal{O}(1) \qquad (\text{pre-agentic: } c_{\mathcal{S}} \gg c_{\mathcal{P}})$$

SOURCES Pitkäranta & Pitkäranta 2026 · Formal sketch (math.html) · Brooks 1995 · Boehm 1981

"Every visualisation above shares one hidden premise: the problem can be substantially understood before any working artefact exists."

FORMAL SKETCH MATH.HTML · §2.1

Eight frames for the discovery side

1 · Domains

$\mathcal{P},\ \mathcal{S}$

candidate needs and artefacts, partly observable

2 · Fit functional

$\varphi : \mathcal{P} \times \mathcal{S} \to [0, 1]$

noisy, estimated locally from use

3 · Joint search

$\arg\max\ \varphi(p, s)\,\pi(p)$

problem and solution searched together

4 · Cost ratio

$c_{\mathcal{S}}/c_{\mathcal{P}} \to \mathcal{O}(1)$

structural, not a productivity claim

5 · Permeable seam

$\nabla_s\,\varphi(p, s)$

gradient from a running probe

6 · Incumbent

$s_0,\ \kappa,\ \rho(s_0; \kappa)$

incumbent, constraints, pain: discovered, not assumed

7 · Value of unlock

$V(p; s_0) = \rho(s_0; \kappa) - \rho(s^*; \kappa)$

how much pain disappears

8 · Two modes

$\pi(p \mid o) \propto L(o \mid p)\,\pi(p)$

validating updates π, proposing updates φ

SOURCES Formal sketch (math.html) Pitkäranta & Pitkäranta 2026

FORMAL SKETCH MATH.HTML · §4.1

Engineering fixes p* and searches for quality s

$$s^* = \arg\max_{s \in \mathcal{S}} \varphi(p^*, s) \qquad (9\text{: spec as fixed input})$$

$$\mathcal{S}_{\text{prod}} = \{ s \in \mathcal{S} \mid Q_j(s) \geq \tau_j \ \ \forall j \in \{1, \ldots, m\} \} \qquad (10)$$

$$s \circ E \ \rightsquigarrow \ E' \qquad (11\text{: integration})$$

$$V(s) = \bigwedge_{k=1}^{n} v_k(s), \ \ v_k(s) \in \{0, 1\} \qquad (12)$$

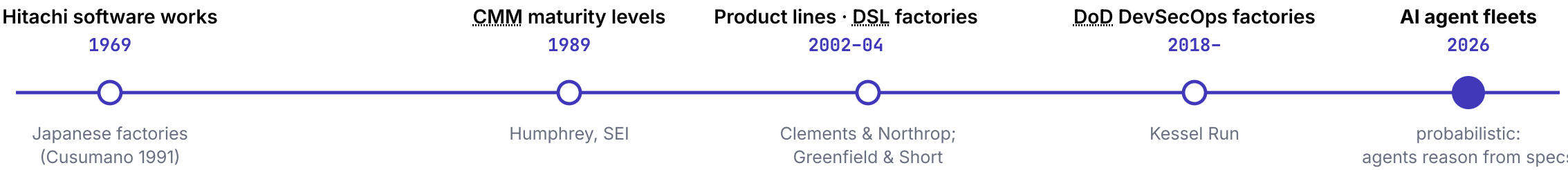
Discovery searches $\mathcal{P} \times \mathcal{S}$ jointly; Engineering fixes p* extracted from a validated probe and searches $\mathcal{S}$ under quality ($Q_j \geq \tau_j$), integration ($s \circ E$) and verification ($V(s)$). Probes deliberately live outside $\mathcal{S}_{\text{prod}}$.

Note: the source uses V for both the value of the unlock $V(p; s_0)$ and the verification functional $V(s)$.

SOURCES Formal sketch (math.html) Pitkäranta & Pitkäranta 2026

SOFTWARE FACTORY LINEAGE 1969–2026

The probabilistic factory and its discovery sibling



Japan's Software Factories
Cusumano 1991

Taxonomy
Aaen et al. 1997

CMM
Humphrey 1989

Product Lines
Clements & Northrop 2002

MS Software Factories
Greenfield & Short 2004

DoD / Kessel Run
Keller et al. 2022 (RAND)

The Factory Model
Osmani 2026

AE is the "probabilistic Software Factory"; AD is its discovery-side sibling, where queue intake, not throughput, binds.

SOURCES Cusumano 1991 · Osmani 2026 · Pitkäranta & Pitkäranta 2026

Figure: talk deck

SECTION

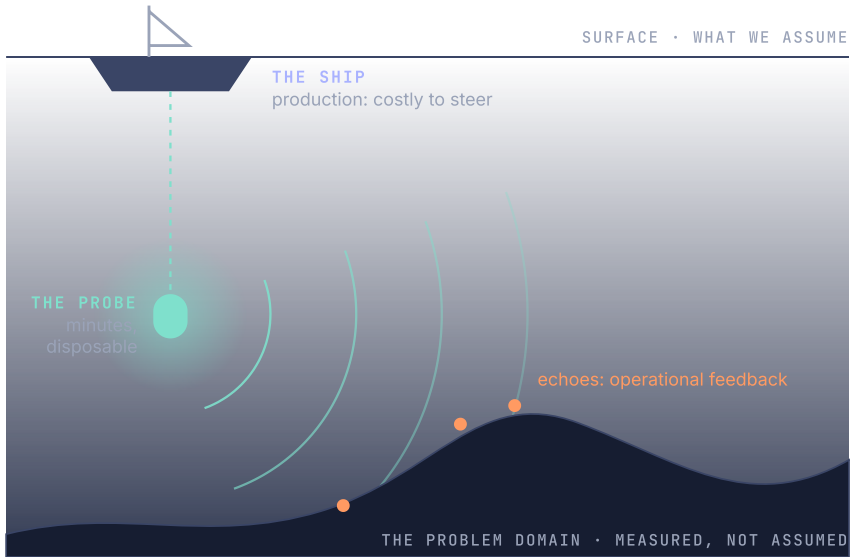
Agentic Probe

The instrument of Agentic Discovery: a runnable, throwaway probe into the problem space, built by an agent from the stakeholder's own words. What it is, why it must be run rather than read, and how it learns about the problem.

SOURCES Pitkäranta & Pitkäranta 2026

DEFINITION PAPER §3.1 · EDITION 2026-10-01

The Probe: a sounding into the problem space



> "A **Probe** is a stakeholder-runnable instance of a candidate solution to a stated pain. It is throwaway-grade by design -- its job is to validate or falsify the pain, not to be production-ready."

Forms

working software · process simulation · walkable future scenario · mixed renderable

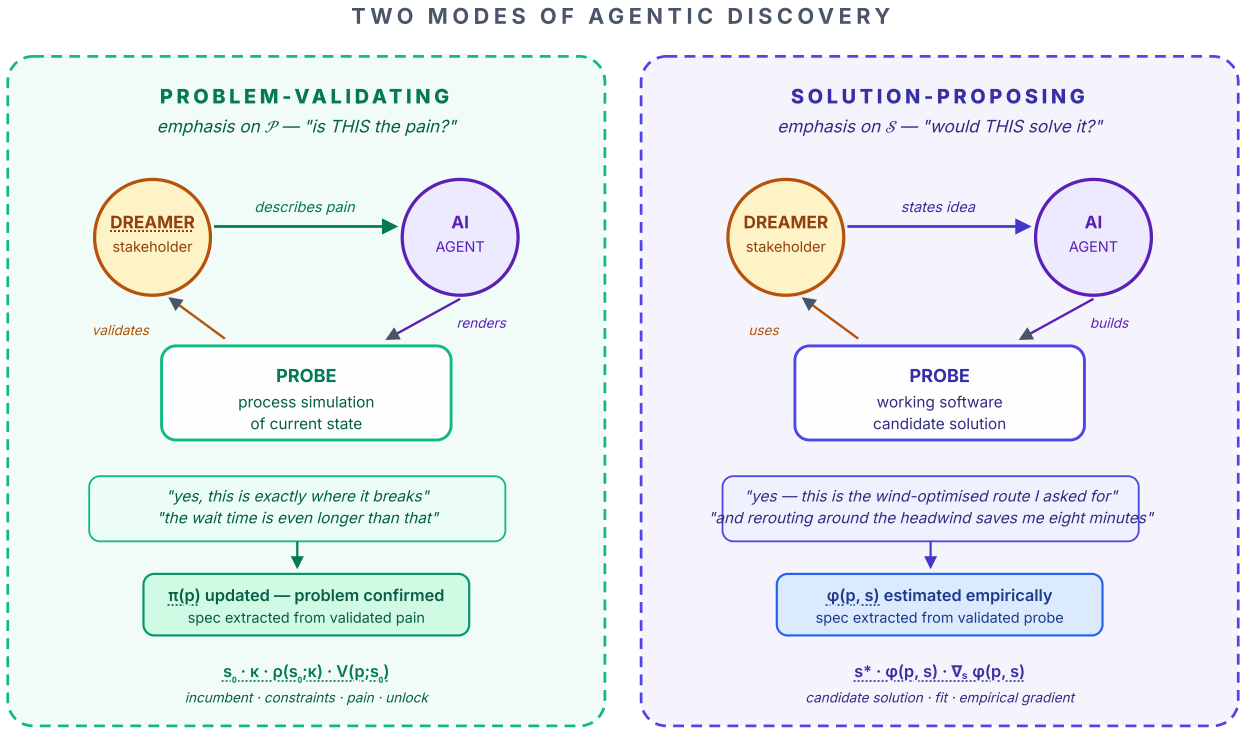
Edition 2026-10

agentic, usable, disposable. A prototype reveals φ, a simulation s_0, κ and ρ, a possible future V.

SOURCES Pitkäranta & Pitkäranta 2026

PAPER FIG. 1 §3.2

Problem-validating and solution-proposing probes

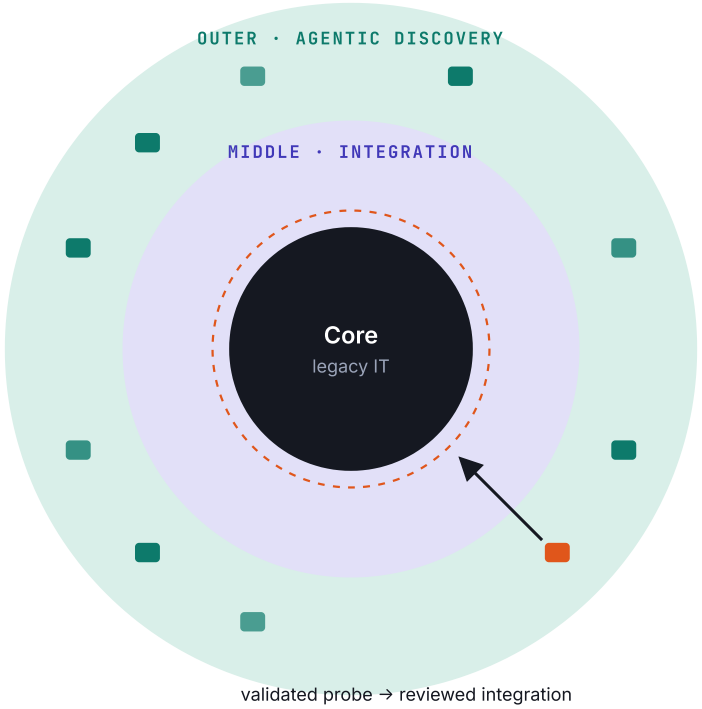


$$(p^*, s^*) = \arg\max_{(p,s)\in\mathcal{P}\times\mathcal{S}} \varphi(p,s)\cdot\pi(p), \qquad \pi(p \mid o) \propto L(o \mid p)\,\pi(p), \qquad \widehat{\nabla_s \varphi} \text{ from probe use}$$

SOURCES Pitkäranta & Pitkäranta 2026 Formal sketch (math.html)

PAPER FIG. 6 §3.3 · §6.8

The probe maps constraints outside-in, inside-out or in between



Outside-in

Start with a new AI process in the outer ring. The probe shows which integrations and core parts are really needed.

Middle-out

Start from the integration layer (APIs, connectors). The probe tests what can change without touching the core.

Inside-out

Interview and reflect the current process as a simulation of the core, then propose paths outward; in regulated settings this eases adoption.

A constraint is a part that cannot or should not change: legacy IT, regulation, data. The probe reveals which constraints are real, i.e. which parts to keep and which to let go. A failed probe costs only in the outer ring.

SOURCES Pitkäranta & Pitkäranta 2026

FORMAL BACKGROUND DENOTATIONAL SEMANTICS

Is any piece of software a function?

Program text is a string; the function is its **meaning** ⟦P⟧. Its shape depends on what the program does:

FEATURE	⟦P⟧
Pure computation	$f : A \to B$
May not terminate	$A \rightharpoonup B$
Mutates state	$\Sigma \times A \to \Sigma \times B$
Interactive (UI, server)	$F : I^\omega \to O^\omega$
Randomness	$A \to \Delta(B)$
Nondeterminism, concurrency	$A \to \wp(B)$
Effects in general	$A \to T(B)$

Yes, once the input is chosen right

A deterministic program is a function once the input includes everything the output depends on: input history, state, clock, network replies and random seed.

Interaction: where is the human?

A stateful system is a pure function in a loop: δ : S × I → S × O yields a stream function F : I^ω → O^ω (Mealy, Kahn 1974). The only contested point is whether the environment, i.e. the human, is inside the function or outside it (Milner 1993, Wegner 1997). Genuine nondeterminism is a relation.

Consequence for the probe

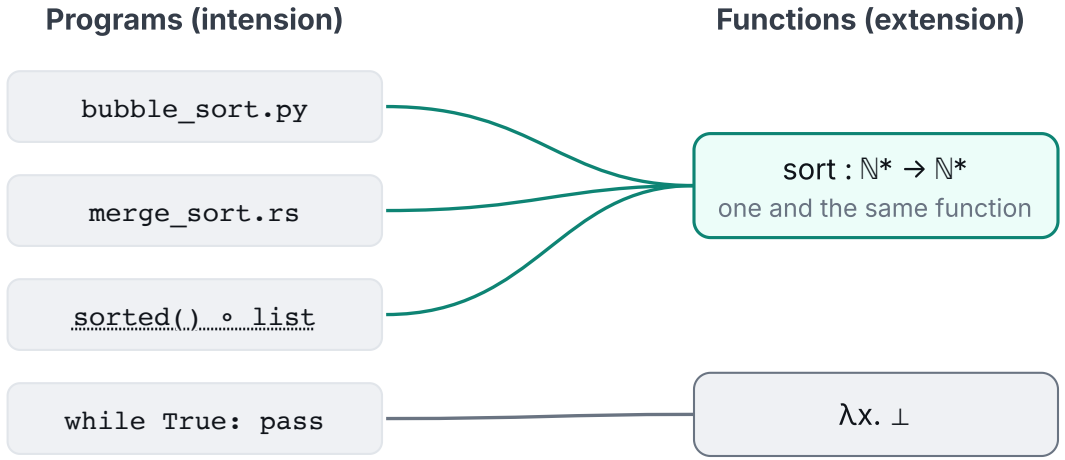
Since every program is a function, “a probe is a function” does not distinguish a probe from any app. A probe is the choice to put the human inside the function.

SOURCES Scott & Strachey 1971 · Moggi 1991 · Kahn 1974 · Milner 1993 · Wegner 1997

Scott & Strachey 1971 (denotational semantics), Moggi 1991 (effects as monads), Kahn 1974 (stream functions), Milner 1993 and Wegner 1997 (interaction).

FORMAL BACKGROUND RICE'S THEOREM

Why a probe must be run, not read



⟦ · ⟧ forgets time, memory and structure

The map P ↦ ⟦P⟧ is many-to-one: a program is a description of a function (intension), the function its extension.

$$\mathcal{C} \text{ non-trivial} \implies \{\, e \mid \varphi_e \in \mathcal{C} \,\} \text{ undecidable}$$

Rice's theorem (1953)

No non-trivial property of a program's behaviour can in general be decided by reading its text. Behaviour has to be run to be seen.

Consequence for the probe

If even a program's own properties require running it, fit φ(p, s), which also depends on a human, cannot be read off a spec. This supports the paper's finding "the cheapest critic is a running probe".

Caveat: Rice concerns properties of programs, not fit to users; for φ this is an analogy, not a proof.

SOURCES Rice 1953 Pitkäranta & Pitkäranta 2026

Rice 1953. Figure adapted from the essay "Is everything a function?" (ohjelmisto-funktiona/). Synthesis: the link to the probe is this deck's.

SYNTHESIS THIS DECK'S FORMULATION, NOT A CLAIM OF THE PAPER

Software maps inputs; a probe maps the problem

Level 1 · Software

The probe artefact is a point $s \in \mathcal{S}$. Its semantics $[\![s]\!] : \mathcal{I}^* \to \mathcal{O}$ is usually deterministic, like any app.

Level 2 · Sounding

Closing the loop with a stakeholder u_p, whose behaviour depends on the hidden problem p, yields a stochastic map from problems to observations: a statistical experiment in Blackwell's sense.

$$\underbrace{[\![s]\!]}_{\text{software}} \circlearrowleft \underbrace{u_p}_{\text{stakeholder}} \implies \mathrm{Probe}_s : \mathcal{P} \to \Delta(\mathcal{O}), \quad p \mapsto L(\cdot \mid p, s)$$

$$b \xrightarrow{A} s \sim A(\cdot \mid b) \xrightarrow{\text{use}} o \sim L(\cdot \mid p^*, s) \xrightarrow{\text{Bayes}} \pi(p \mid o) \propto L(o \mid p, s)\,\pi(p)$$

Randomness comes from the human

The finished probe code is deterministic; the sounding function exists only because the Dreamer is in the loop, which grounds the role formally.

Two modes = two $\mathcal{O}$s

Validating: observations of s_0, κ, ρ (updates π). Proposing: samples of φ(p,s) and $\nabla_s \varphi$.

"Throwaway" made precise

A probe is judged by informativeness I(p; o | s), not production quality. Choosing a probe is Bayesian experimental design: max I(p; o | s) / $c_{\mathcal{S}}$.

SOURCES Pitkäranta & Pitkäranta 2026 · Blackwell 1953 · Lindley 1956 · Formal sketch (math.html)

b = brief, A = agent, itself stochastic: the same brief does not yield the same probe. The paper's L(o | p) is written L(o | p, s).

LITERATURE REVIEW · OCTOBER 2026

Agentic Search & Agentic Information Retrieval

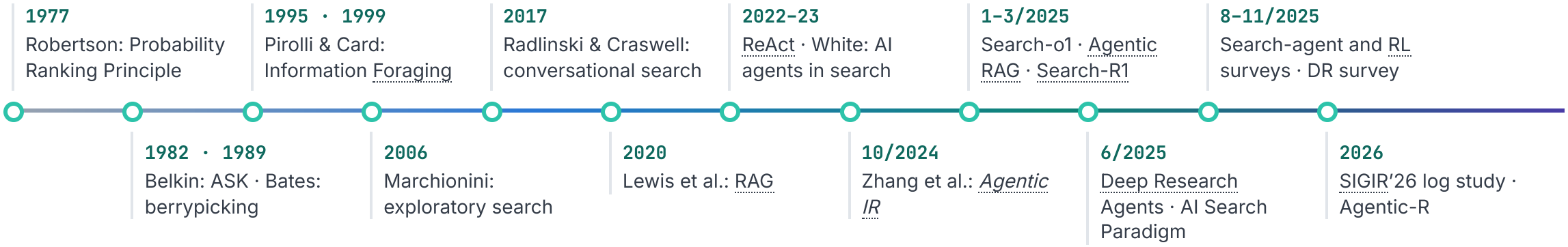
How the concept is defined in the academic literature: definitions, formal models and taxonomies from about 35 sources.

Zhang et al. 2024 · Huang et al. 2025 · Xi et al. 2025 · Lin et al. 2025 · Shi et al. 2025 · Search-R1 · ReAct · …

SOURCES Zhang et al. 2024 (Agentic IR) Huang et al. 2025 (Deep Research Agents)

TIMELINE

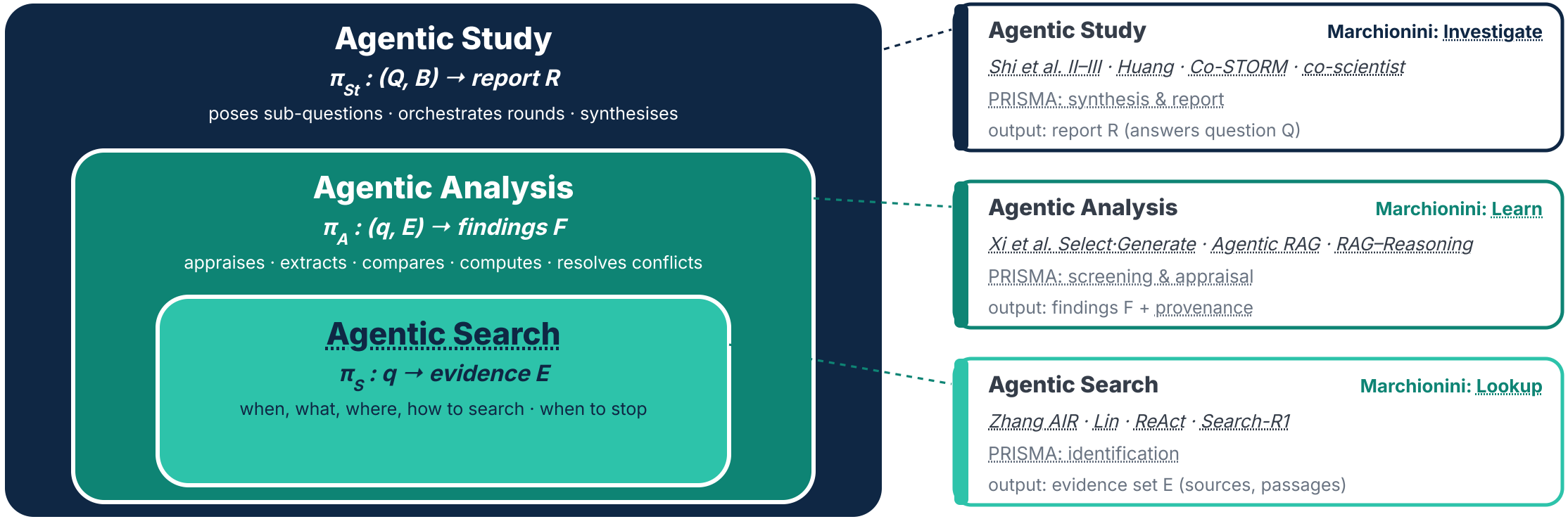
Half a century of search: from ranking to agents



SOURCES Robertson 1977 · Belkin et al. 1982 · Bates 1989 · Pirolli & Card 1999 · Marchionini 2006 · Radlinski & Craswell 2017 · Lewis et al. 2020 (RAG) · Yao et al. 2023 (ReAct) · White 2023 · Zhang et al. 2024 (Agentic IR) · X. Li et al. 2025 (Search-o1) · Jin et al. 2025 (Search-R1) · Ning et al. 2026

CONCEPT MAP THIS DECK'S SYNTHESIS

Agentic Study ⊃ Agentic Analysis ⊃ Agentic Search



Each outer layer calls the inner one as a subroutine and decides itself when to call it again. The layers match Marchionini's (2006) three search activities and the PRISMA 2020 review stages; the layering is our synthesis, not any single source's terminology.

SOURCES Marchionini 2006 · Page et al. 2021 (PRISMA 2020) · Shi et al. 2025 (Deep Research survey) · Xi et al. 2025 (search agents) · Zhang et al. 2024 (Agentic IR) · Huang et al. 2025 (Deep Research Agents)

DEFINITIONS THE LAYERS MAPPED TO THE LITERATURE

Three nested policies, each with its own output

$$E = \pi_S(q), \qquad F = \pi_A\big(q, E\big), \qquad R = \pi_{St}(Q, B) = \mathrm{synth}\Big(\{\pi_A(q_i, \pi_S(q_i))\}_{i=1}^{n}\Big)$$

LAYER	DEFINITION	OUTPUT	MARCHIONINI 2006	SHI ET AL. 2025	XI ET AL. 2025 LOOP	PRISMA 2020
Agentic Search	Decides over many steps when, what, where and how to search, and when to stop.	evidence E	Lookup	I · Agentic Search	Plan · Act · Reflect	identification
Agentic Analysis	Appraises, extracts, compares and combines search results; may search again if evidence is insufficient.	findings F + provenance	Learn	I "extracting" + II synthesis	Select · Generate	screening, appraisal
Agentic Study	Poses sub-questions q_i, runs many search and analysis rounds within budget B, synthesises and reports.	report R	Investigate	II Integrated Research · III AI Scientist	the whole loop, repeated	synthesis, reporting

Boundaries. Search ↔ Analysis: does a step change the evidence set E (search) or the interpretation F (analysis)? Analysis ↔ Study: is the question given (analysis) or does the agent pose its own sub-questions and report (study)? Shi et al. do not separate analysis as a phase; that is this synthesis's addition.

SOURCES Marchionini 2006 · Shi et al. 2025 (Deep Research survey) · Xi et al. 2025 (search agents) · Page et al. 2021 (PRISMA 2020) · Huang et al. 2025 (Deep Research Agents) · Y. Li et al. 2025 (RAG–Reasoning)

STARTING POINT

One word, four different things

AGENTIC RAG

Agent in the pipeline

The model decides *when* to retrieve. Retrieval is still a single RAG step.

AGENTIC SEARCH

Search as an action

Multi-step, planned, reflective search over many sources. Optimized with RL.

DEEP RESEARCH

Research agent

Long horizon, browser, code and MCP. The output is an analytical report.

AGENTIC IR

A new paradigm

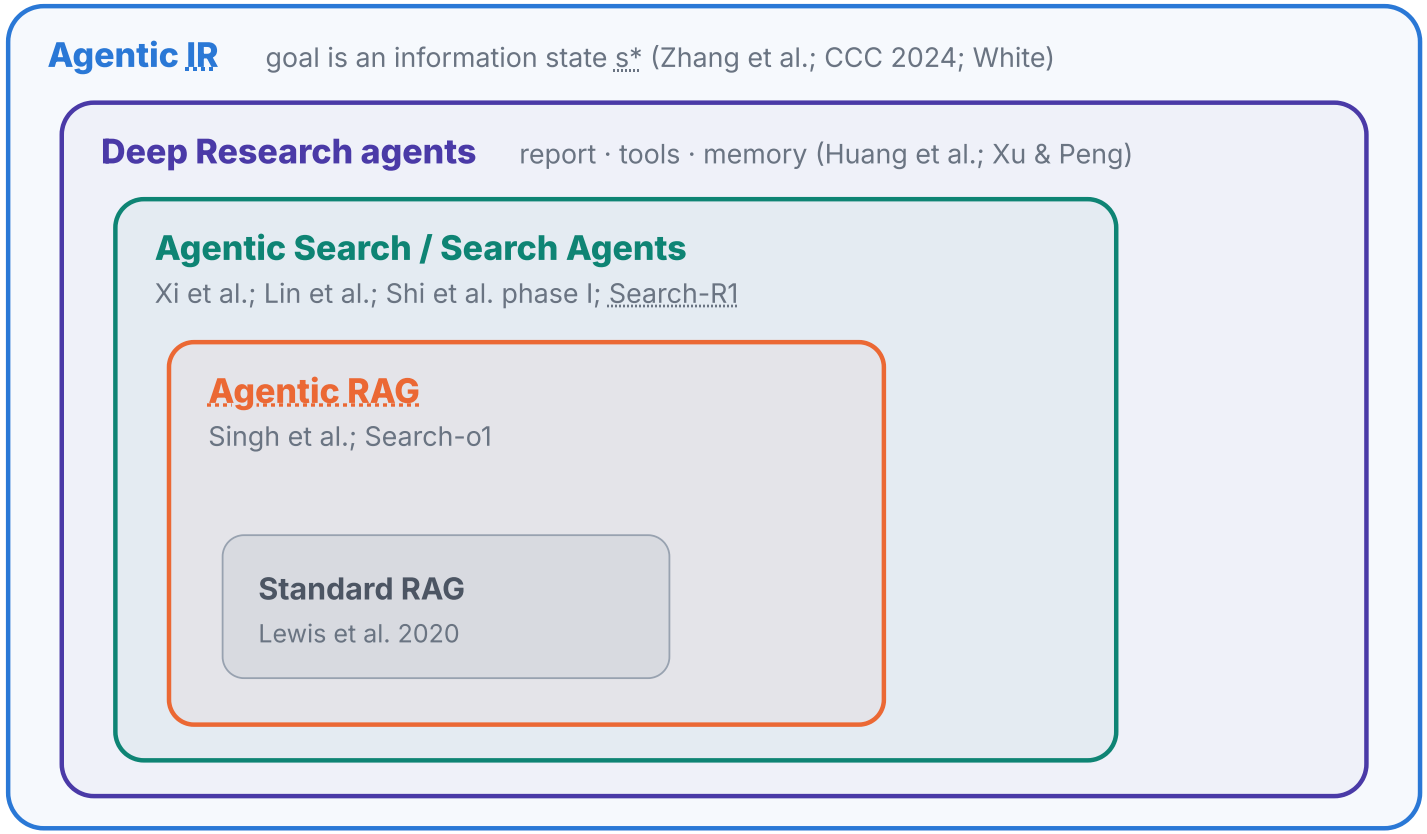
The goal is an *information state*, not a document. Classic IR is a special case.

The terms are used partly as synonyms and partly hierarchically. There is no settled shared definition yet, but the literature shows a **nested structure** and a **common formal core**.

SOURCES Singh et al. 2025 (Agentic RAG) · Xi et al. 2025 (search agents) · Huang et al. 2025 (Deep Research Agents) · Zhang et al. 2024 (Agentic IR)

CONCEPT MAP

Nested concepts and the inverted perspective



Agent as the IR user

- IR for AGI
  Zhai, SIGIR'25
- Agentic Search in the Wild
  Ning et al., SIGIR'26
- Agentic-R
  Liu et al., ACL Findings'26
- Direct Corpus Interaction
  Li et al. 2026

→ retrievers and interfaces are redesigned for agent consumers

SOURCES

Zhang et al. 2024 (Agentic IR) · Huang et al. 2025 (Deep Research Agents) · Xi et al. 2025 (search agents) · Lin et al. 2025 (RL agentic search) · Shi et al. 2025 (Deep Research survey) · Singh et al. 2025 (Agentic RAG) · Lewis et al. 2020 (RAG) · Zhai 2025 · Ning et al. 2026 · Liu et al. 2026 (Agentic-R) · Z. Li et al. 2026 (DCI) · Allan et al. 2024 (CCC) · White 2023

SECTION

Agentic Discovery: experiments

What we did and what we measured: the research design (DSRM), the session protocol, eight field engagements with two organisations (Experiment A, 69 probes) and process simulations in healthcare (Experiment B).

SOURCES Pitkäranta & Pitkäranta 2026 Appendix A.1–A.12

DSRM · IS DESIGN THEORY §1.1 · §6.7

Design science: artefact, objectives O1–O6 and limitations

OBJECTIVE	RESULT
O1 time to first probe	minutes; ≤ 1 h every session
O2 deployability	69/69
O3 spec extractability	n = 9, 1:56
O4 re-engagement	7/8 sites
O5 safety	0 incidents
O6 scalability	8 engagements

DSRM's six activities map to the paper's sections; the two-mode pattern is a new design theory whose kernel theory is the Bayesian update of problem-domain knowledge $\pi(p \mid o)$.

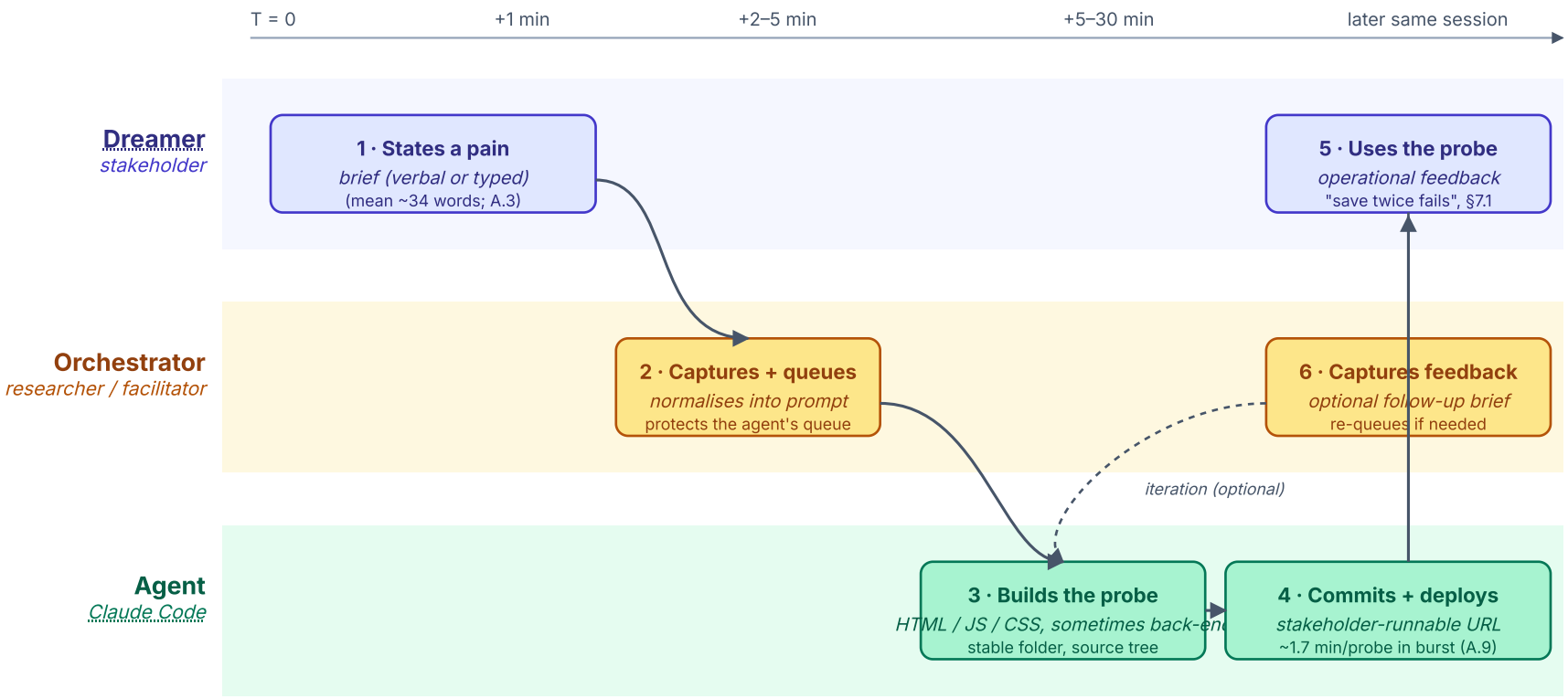
Limitations L1–L6

sample & self-selection · throwaway quality · single vendor · effort baseline · researcher as orchestrator · DSRM scope

SOURCES Peffers et al. 2007 · Hevner et al. 2004 · Walls et al. 1992 · Markus et al. 2002 · Wohlin et al. 2012

PAPER FIG. 9 §5

From pain to runnable probe in minutes

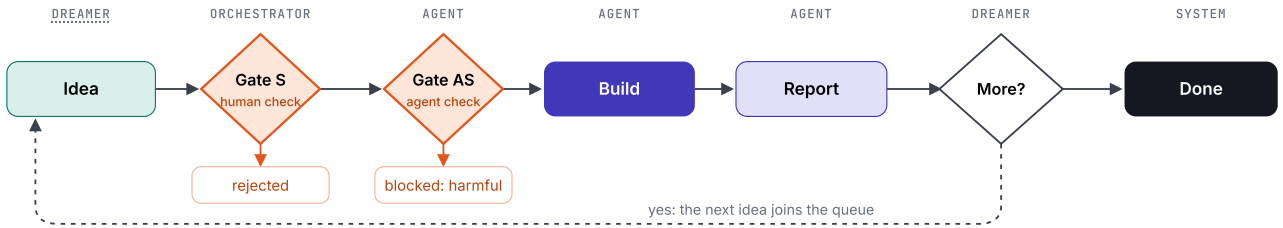


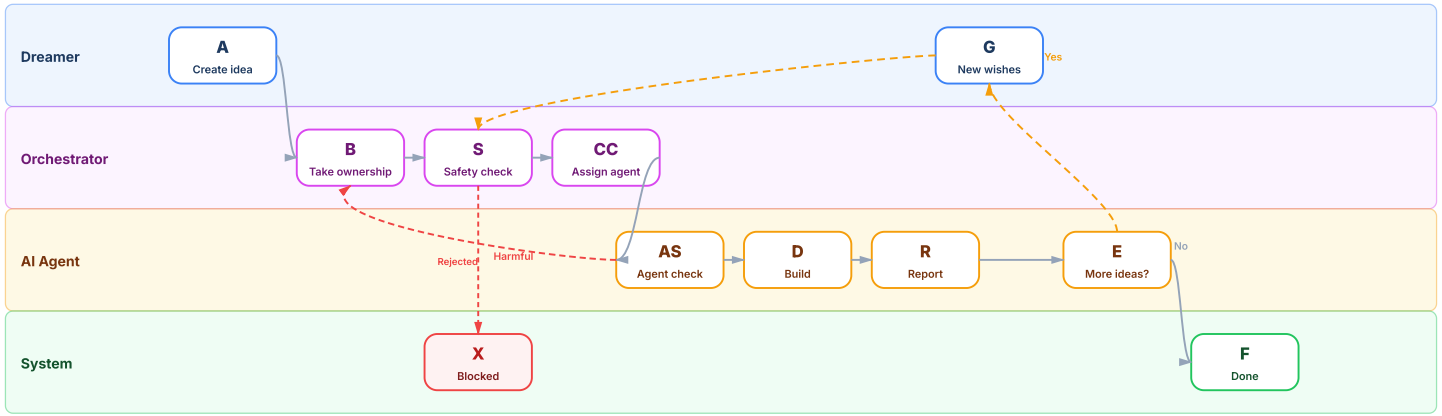
SOURCES Pitkäranta & Pitkäranta 2026 Appendix A.1–A.12

Brief ~34 words on average; ~1.7 min per probe in the burst

PAPER FIG. 8 §3.4 · §5

Dreamer, Orchestrator, Agent and two safety gates



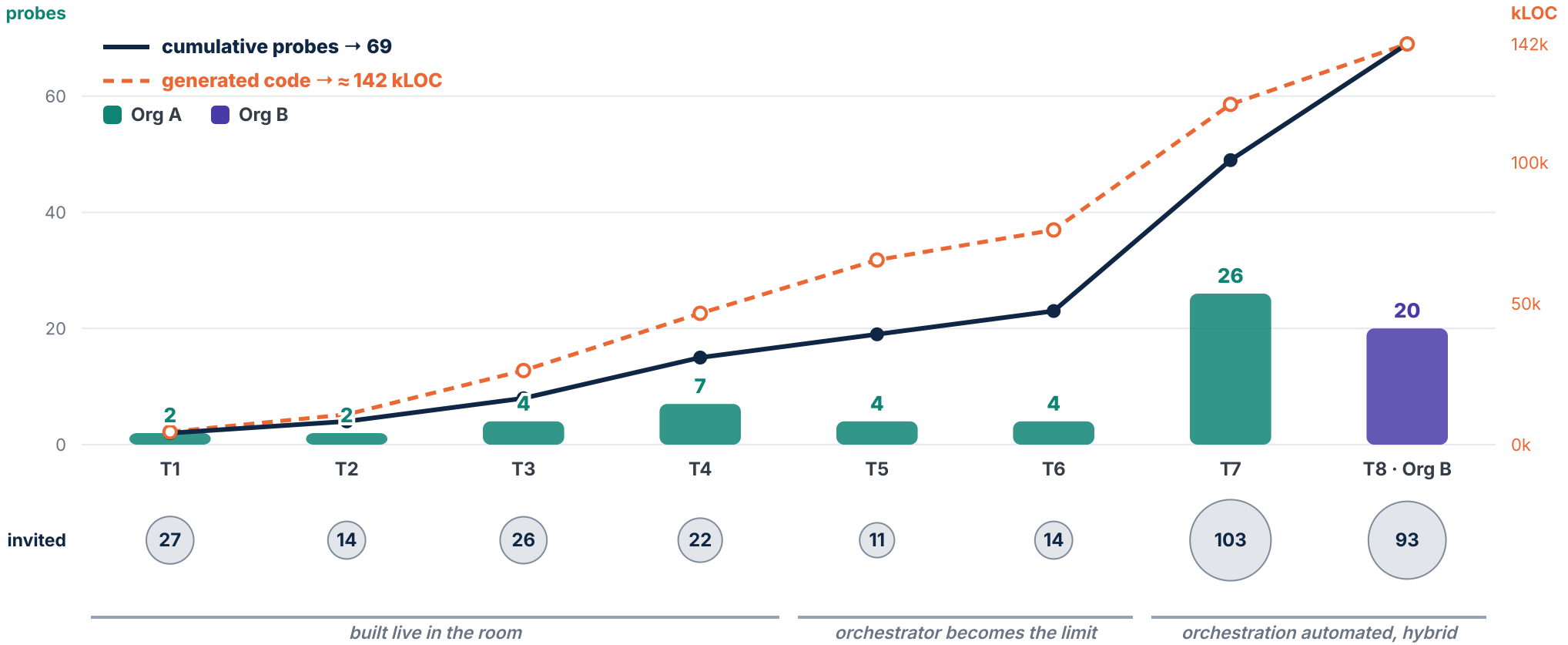


Dreamer: stakeholder, natural language or voice · **Orchestrator**: shapes the prompt, safety gate S, curates (skipping it is an anti-pattern) · **AI Agent**: self-check (gate AS, AGENT_SAFETY.md), build, deploy, report. 0 safety incidents in 69 generations.

SOURCES Pitkäranta & Pitkäranta 2026 Pitkäranta & Pitkäranta 2025

EXPERIMENT A · 8 ENGAGEMENTS IN CALENDAR ORDER · §6.1 · §7.4

69 probes and ~142,000 lines of code in four weeks



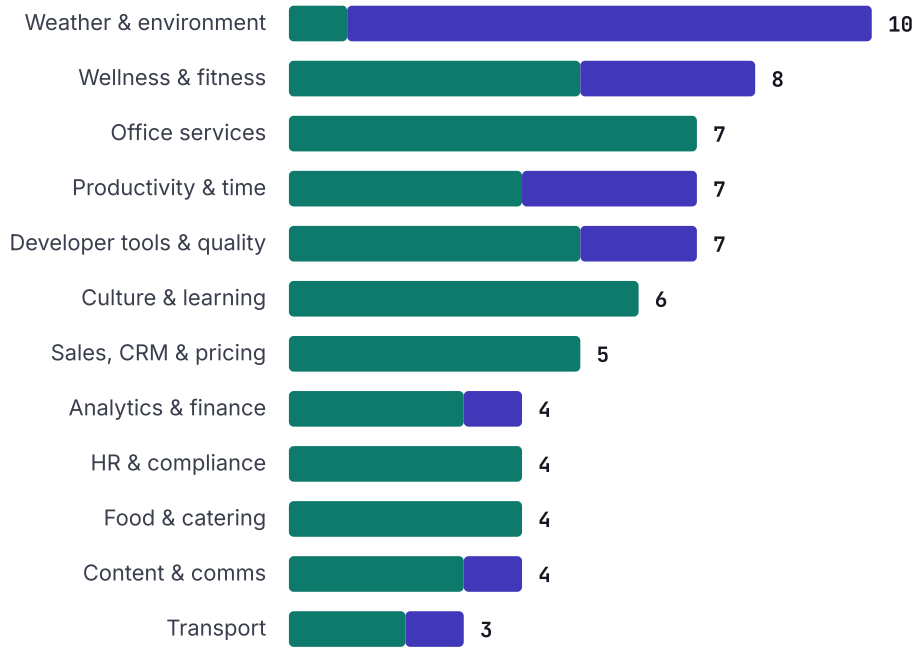
Bar = probes per engagement, solid line = cumulative probes, dashed line = cumulative generated code (kLOC). Circle = people invited (≈ 310 in total; invited is not attended: about 50 were in the room at T8). The two largest engagements produced 46 of the 69 probes.

SOURCES Pitkäranta & Pitkäranta 2026 · Appendix A.1–A.12

Sources: paper §6.1 (probes, LOC) and event log; engagements anonymised

EXPERIMENT A §6.1–6.4 · APPENDIX A.1–A.12

69 probes, 2 organisations, ~118,000 LOC



	ORG A	ORG B	TOTAL
Probes	49	20	69
LOC	~97 500	~20 900	~118 400
Spec lines	~1 430	~540	~1 970
Ratio	1:68	1:39	1:60
Span	~4 wk	1 day	

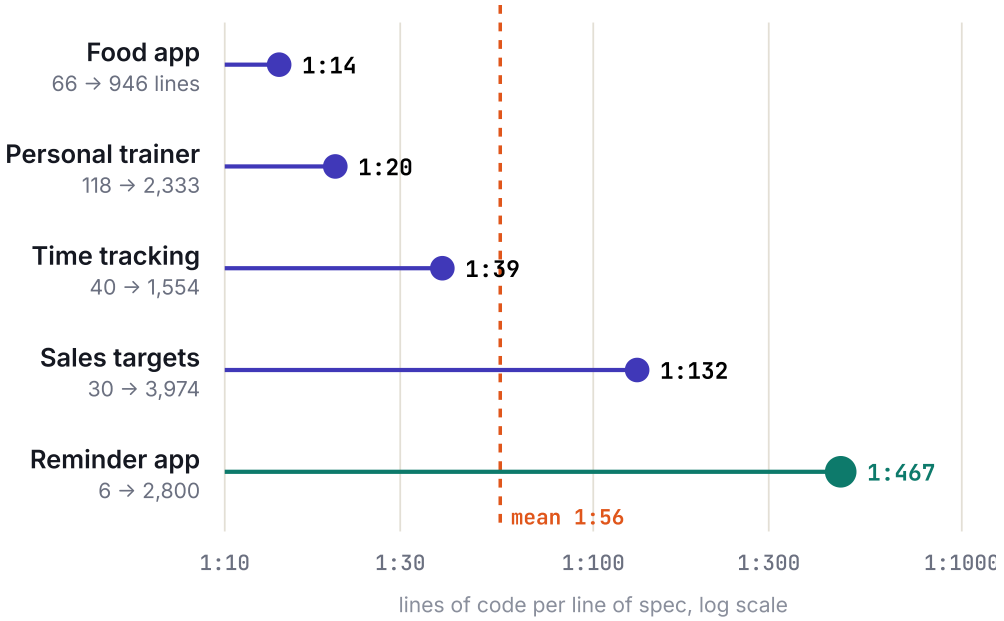
Categories: weather 10, wellness 8, office services 7, productivity 7, dev tools 7… Weather is 45% of Org B. LOC ~142,000 counted at site level vs ~118,400 at cohort level.

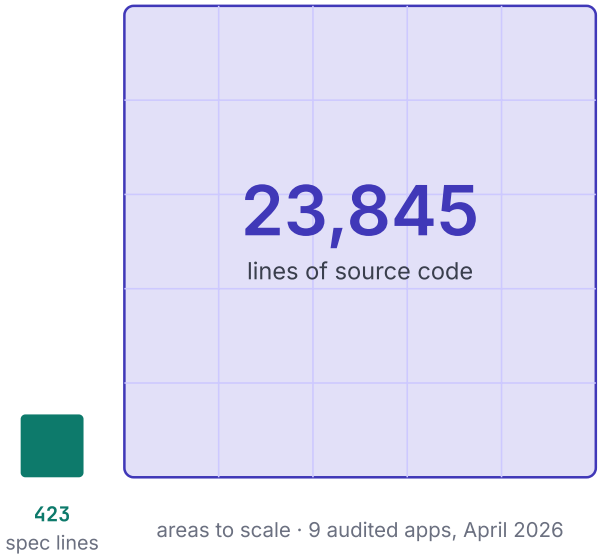
SOURCES Pitkäranta & Pitkäranta 2026 Appendix A.1–A.12

Figure: talk deck · Numbers: paper §6 and appendix A

AMPLIFICATION §6.2 · N = 9

423 spec lines → 23,845 lines of code





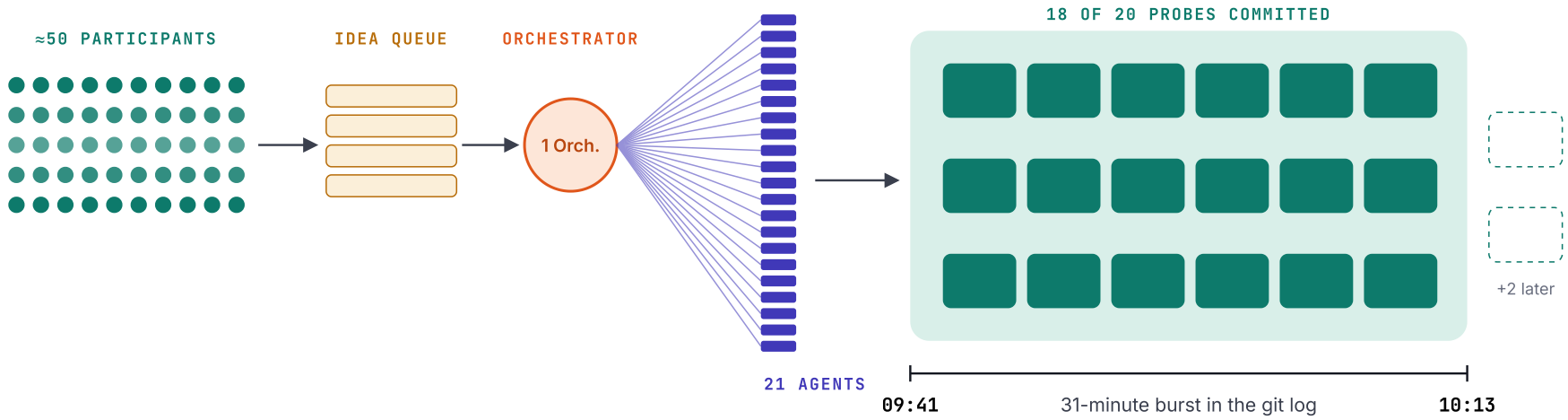
Mean 1:56, median 1:39. A higher ratio reflects a shorter prompt, not more output (appendix A.6).

SOURCES Pitkäranta & Pitkäranta 2026 Appendix A.1–A.12

Figures: talk deck

BOTTLENECK SHIFT C5 · §6.5 · §7.4

From production to orchestration to idea supply



Events 1–4
2–4 parallel requests, all built live: production capacity binds.

Event 5
7 requests: the Orchestrator becomes the bottleneck.

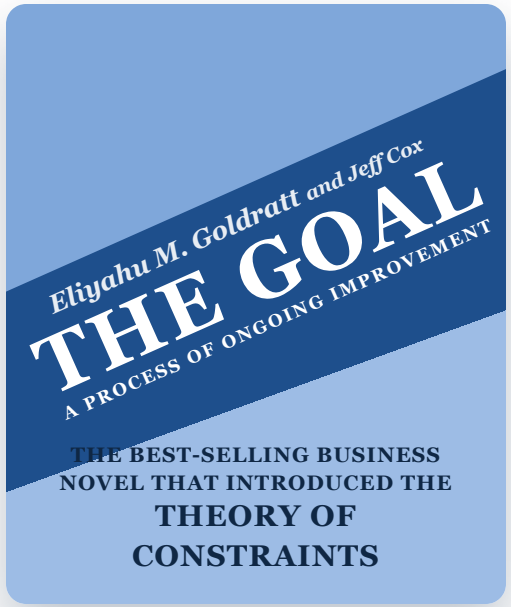
Largest
~100 invited; Org B 18/20 probes in 31 min: idea supply and validation bind.

Speed multiplier ~570× on work time (37–65× with a 10× quality discount); ~2,370 person-days traditional vs ~33 h of prompt writing.

SOURCES Pitkäranta & Pitkäranta 2026 | Appendix A.1–A.12 | Osmani 2026

Figure: talk deck

The bottleneck moves, but the goal stays the same



About the book ↗

Five steps ↗

Goldratt & Cox, North River Press, 1st edition 1984. A business novel about plant manager Alex Rogo, who saves his plant one bottleneck at a time. Over 10 million copies sold, per the cover.

Agents do not remove the constraint; they move it. Goldratt's five focusing steps show why:

1 Identify
Before agents the constraint was production: writing code set the pace, and the whole SDLC was built around it.

2 Exploit
Agents make building cheap and fast ($c_{\mathcal{S}}/c_{\mathcal{P}} \to \mathcal{O}(1)$). Code is no longer what everyone waits for.

3 Subordinate
The rest of the process is paced by the new constraint: orchestration and human attention now set the tempo.

4 Elevate
The constraint is widened by automating orchestration and giving agents more autonomy within the gates.

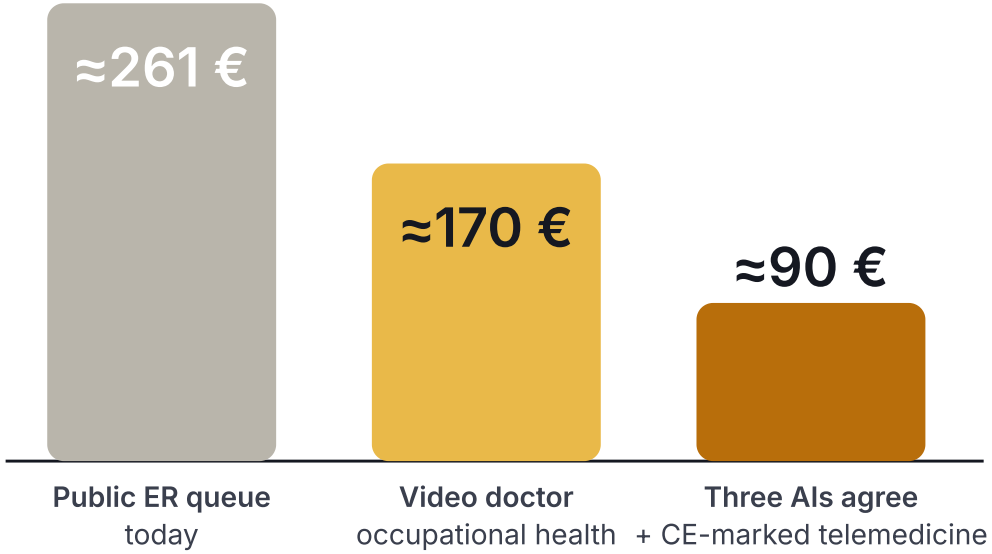
5 Repeat
The bottleneck moves upstream: to finding the right problem and validating it. That is why the agentic SDLC leans on Discovery.

SOURCES Goldratt & Cox 1984 · Pitkäranta & Pitkäranta 2026

The reading of the steps is this deck's synthesis; the field study showed the same path: production → orchestration → ideas and validation (paper C5, §7.4).

PROBLEM-VALIDATING MODE §6.8 · N = 3

Relaxing κ reveals V



$$V(p; s_0) = \rho(s_0; \kappa) - \rho(s^*; \kappa)$$

Three researcher-authored cases: prescription renewal, child illness before a trip, Friday skin symptom. Scenario 1 renders s_0, κ, ρ; scenarios 2–3 relax κ (regulation, billing, locality, physical visit).

Not a field study; the mode is tested only at companion scale (limitation L6).

SOURCES Pitkäranta & Pitkäranta 2026

Figure: talk deck · example: healthcare process discovery companion

SECTION

Summary

The deck in two slides: what we claim, on what evidence, and what it means for product development.

SUMMARY 1/2 SITUATION · CLAIM · INSTRUMENT · EVIDENCE · ENGINE

Building got cheap, so finding the right problem is the new bottleneck

1 Where we are

Capability keeps rising, and faster: METR's time horizon passed its 16-hour ceiling in spring 2026 and the yardsticks are running out. The environment and harness matter almost as much as the model, and coding agents are everyday tools.

2 Core claim

When building gets cheap ($c_{\mathcal{S}}/c_{\mathcal{P}} \to \mathcal{O}(1)$), the six-phase SDLC collapses into two: **Agentic Discovery** finds the right problem (learning velocity), **Agentic Engineering** builds it right (production reliability). The seam is the validated spec.

3 Instrument

The **agentic probe** is a runnable, throwaway instrument into the problem space: problem-validating or solution-proposing. It must be run, not read, and it also reveals the constraints.

4 Evidence

8 field engagements, 2 organisations, 69 probes and ~142,000 lines of code in four weeks. Intent-to-code 1 : 56 on average (up to 1 : 467). The bottleneck moved from production to orchestration, then to idea supply and validation.

5 Engine

Discovery is agentic search over $\mathcal{P} \times \mathcal{S}$. Agentic Search ⊂ Analysis ⊂ Study; the human acts as an ask-or-act gate, and autonomy is a design choice, not a capability.

SOURCES METR 2026b · Pitkäranta & Pitkäranta 2026 · Appendix A.1–A.12 · Zhang et al. 2024 (Agentic IR)

SUMMARY 2/2 PRACTICES · IMPLICATIONS · LIMITS

Agents do the production; humans choose the problem and validate

Market & customer

LLMs can stand in for human groups and produce demand curves, but their biases are systematic. The best results come from humans and LLMs together.

Requirements & spec

Simulated users surface latent needs, but tacit knowledge mostly stays hidden from the agent. The human is formally necessary: they validate the requirements and the domain knowledge.

What an organisation should do

Once production is no longer the limit, invest in stakeholder time, orchestration and validation gates. Build probes to learn, not to ship, and hand the spec to Engineering only once validated.

Limits and open questions

Two voluntary organisations and one agent vendor. The problem-validating mode was tested only on a small scale (n = 3). A probe is not production code, and the speed multiplier does not carry straight over to delivery.

Condenses the slides before the appendix. Claims and numbers come from the preceding sections; the organisational advice is this deck's synthesis.

SOURCES Argyle et al. 2023 · Brand et al. 2023 · Pitkäranta & Pitkäranta 2026